

From Non-Guarantee to Answerability

Cross-Domain Stress Tests of a Universal Formal Kernel

1. Introduction

A formal method intended for use across scientific domains faces a basic design problem. If the shared machinery contains too much domain-specific structure, its apparent generality may disappear as soon as the scientific object changes. If it contains too little, unchanged reuse may be trivial: every difficult scientific distinction can simply be pushed into an application-specific layer while the shared formal core contributes little more than notation. A third failure arises when formalization acquires authority it has not earned, so that machine closure is mistaken for empirical truth or a successful translation is mistaken for scientific identification. A fourth appears when the formal architecture guarantees the kind of outcome the inquiry was supposed to determine.

The Structural Flow Universal Kernel was developed under the opposite constraint. Its starting posture is non-guarantee: neither a preferred scientific outcome nor stronger identity, mechanism, persistence, closure, or downstream conclusion is supplied merely because an inquiry would benefit from it. The universal layer is intended to carry only reusable structural burdens that survive that restriction. Domain-specific science remains responsible for the mechanisms, observations, measurements, comparators, empirical predicates, and evidentiary judgments required to apply those burdens to a particular problem.

This architecture creates an immediate methodological question. How far can a shared formal system remain useful if it does not receive the scientific answer in advance?

The present paper examines that question through a frozen cross-domain corpus. Eleven source-audited scientific application packages had been developed far enough to permit comparative formal analysis before the corpus was frozen. They address failure-regime discrimination, consciousness, quantum gravity, dark energy, the chemistry-to-life boundary, quantum measurement, evolutionary individuality, the thermodynamic arrow, baryogenesis, turbulence, and cosmological-constant mechanisms. A twelfth case in morphogenesis and regeneration is retained separately because it reached substantial causal experimental architecture but was deliberately terminated before protocol freeze and vertebrate execution. The corpus is heterogeneous, but it is not random, statistically representative, or

independently replicated. It is a within-program stress sample of applications developed before the present cross-application claims were fixed.

The simplest result of the audit is also the central historical observation of the paper:

local architecture changed substantially; the shared kernel did not.

Across the eleven completed source-audited applications, no case demonstrated a reusable, domain-neutral structural requirement that required amendment of the frozen Universal Kernel at the application's declared scope. The morphogenesis stress case likewise reached substantial experimental architecture without exposing a kernel-repair burden before the program was stopped. This non-revision result is publicly checkable at the file-identity level because the same kernel file was carried in the archived machine-backed packages and its archive-recorded checksum remained unchanged.

File stability alone would be weak evidence. The analysis therefore asks what that stability means. Some applications accumulated extensive local formal structure, specialized evidence grammars, comparator classes, experimental controls, and result-routing rules. Others directly exercised deeper reusable structures in the kernel beyond the common translation interface. Direct theorem-family coverage nevertheless remained uneven. The paper therefore does not treat repeated inclusion of one file as equivalent to exhaustive testing of that file, nor does it treat local complexity as evidence that the universal layer somehow contains the relevant domain science.

The more informative result is architectural. Across the frozen corpus, substantial scientific complexity could remain owned by the scientific layer while a smaller shared formal layer continued to provide common admission, translation, lineage, answerability, failure, openness, and selected closure constraints. The applications varied in scientific object, evidentiary posture, local formal complexity, and lawful result structure without requiring those differences to be promoted automatically into universal machinery.

This separation of authority is essential to the claim. The kernel does not infer whether a physical mechanism is real, whether a biological assay is adequate, whether a comparator class is scientifically exhaustive, or whether an empirical observation has been correctly interpreted. Those judgments remain domain-owned. An adapter or contract represents the scientifically warranted state at a declared object, level, and scope; the machine checks conditional consequences of that encoded record. A machine certificate can therefore establish formal validity relative to a faithfully populated record without establishing either the faithfulness of the translation or the empirical truth of its premises.

The corpus also tests whether non-guarantee merely produces weak inquiry. It did not do so in these applications. The lawful outcomes include bounded positive conclusions, negative results, exact non-identification, unresolved burdens, explicit blockers, composite results, and multiple admissible successors. In several cases, the formal architecture makes a stronger conclusion unavailable precisely because one required burden remains open. In others, substantial closure is licensed at one scope while an adjacent scientific question remains unanswered. Across the frozen corpus, strong scoped closure or empirically actionable answerability did not require replacing the universal non-guarantee posture with a universal premise that preselected the scientific outcome.

“Answerability” is used here in that bounded sense. It does not mean that every open scientific problem has been solved, or even that every application has reached empirical adjudication. It means that an inquiry has been narrowed enough for the relevant burden, admissible evidence, failure condition, unresolved dependency, or lawful conclusion to become explicit. A located OPEN state can therefore represent methodological progress where the alternative is an undifferentiated question whose hidden dependencies remain unexamined.

A recurring feature of the corpus is that broad scientific problem labels often became narrower scientific objects before formal adjudication. The chemistry-to-life work separates continuity through turnover, autonomous maintenance, and the later Life translation. The quantum-gravity program separates quantum description, load-bearing quantum structure, gravity attribution, and comparison-class exhaustion. The cosmological-constant program separates protection, empirical identification, and numerical realization. Similar narrowing occurs elsewhere in the corpus. These repetitions suggest that object disambiguation may be important to preserving the boundary between reusable structure and local scientific burden. The present study does not establish that these decompositions are unique, objectively privileged, or reproducible by independent investigators, and it does not claim that Structural Flow uniquely discovers the correct scientific object.

The paper therefore treats kernel non-revision as something to explain and attack rather than as proof of universality. The adversarial analysis asks whether adapter elasticity makes the result trivial, whether the kernel constrains too little to fail, whether machine soundness has been confused with semantic or empirical authority, whether object formation is circular, whether theorem-family coverage has been overstated, and what would count as genuine revision pressure rather than an ordinary scientific failure or malformed translation. Two direct falsification routes follow: a faithfully populated required burden that the kernel wrongly licenses as conforming, or a coherent scientific object that requires a reusable

domain-neutral structural relation the current kernel cannot represent without distortion or lawful localization below the universal layer.

The present evidence does not establish kernel completeness, minimality, uniqueness, optimality, permanent immutability, empirical correctness of the application programs, or generalization to science as a population. Independent external replication is not yet available. In particular, no independent specialist has yet selected a new problem, constructed the object and translation independently, and subjected the frozen kernel to a full external application. That remains the principal unresolved validation burden.

The contribution of this paper is therefore narrower than a claim that one formal system has solved a collection of open scientific problems. It is a source-audited record of what happened when a frozen, non-guaranteeing formal interface was reused across a heterogeneous within-program corpus: which structure remained shared, where the scientific complexity accumulated, which kernel components were actually exercised, which result forms remained lawful, and what future evidence would require the universal layer to change.

Section 2 defines the architecture and its authority boundary. Section 3 fixes the audit design and frozen corpus. Sections 4–6 examine kernel non-revision, the location of scientific complexity, and the relation between non-guarantee and lawful answerability. Sections 7 and 8 turn the strongest objections into explicit limitations and falsification routes. Section 9 interprets the resulting stability without promoting it beyond the evidence. The conclusion returns to the narrower historical claim: the applications changed; within the frozen audited corpus, the shared kernel did not.

2. Architecture and Authority Boundary

The Universal Kernel is not intended to contain the scientific content of the domains to which it is applied. Its role is narrower: to provide a reusable formal surface on which structurally comparable burdens can be stated, adjudicated, and carried across applications without transferring authority over the underlying science to the formal system. This distinction is central to the architecture. A domain may supply mechanisms, equations, observables, empirical predicates, classification rules, or experimental evidence; the kernel may receive and constrain a translation of those materials, but their scientific warrant remains with the domain that owns them. The governing interface states this explicitly: structural derivation, domain identification, and empirical confirmation carry separate status.

The kernel is therefore best understood as an interoperability and adjudication layer rather than a universal scientific model. Its common translation interface requires every application

to identify what is being translated, at what level and scope, and under what preservation and non-circularity conditions. The minimum admission surface contains eight recurring burdens: the translation surface must itself be independently warranted; the relevant source or carrier must be pinned; the formal object must be identified; level and scope must be fixed; the relevant structural job must survive the translation; the desired conclusion may not be used to establish its own premises; and successful translation may not be inflated into identity, mechanism equivalence, empirical truth, or broader closure.

These burdens are implemented directly in the released Lean kernel. The Universal Translation Contract represents them as a fixed core while allowing each application to register additional object-specific and domain-specific burdens. Domain burdens are further separated into those required for entry into the translation and those that remain to be answered after structural conformance. This is an important asymmetry. A scientific fact may be necessary before a formal translation is even meaningful, while another equally important scientific fact may remain unresolved after the structural relation has already been established. The latter does not become universal merely because the formal system can locate it.

The resulting formal vocabulary is deliberately small. A registered burden may be discharged, violated, or open. A required violation blocks lawful translation. A required unresolved burden blocks structural conformance without being converted into failure. Structural conformance is available only when the required universal, object-specific, and domain-entry burdens have been discharged. Downstream scientific burdens may nevertheless remain open or later return a negative result without retroactively changing the earlier formal verdict unless that result exposes a contradiction in an entry condition or higher dependency. The corresponding Lean definitions implement the same gating distinction: every required core, object, and domain-entry burden must be discharged for **StructuralConforms**, while known failures and known open burdens receive separate formal treatment.¹

This yields three different kinds of authority that should not be collapsed.

The first is **scientific or empirical authority**. The relevant scientific community and evidentiary record determine whether a mechanism, measurement, comparator, experimental condition, or domain predicate is warranted. Lean does not decide these facts from raw data or from formal resemblance.

¹ $\neg(\mathcal{V})/\neg$ — A Derivation from Non-Guarantee, Second Edition, v1.5, Zenodo, <https://doi.org/10.5281/zenodo.22736388>. The executable formal object referenced here is `Structural_Flow_Universal_Kernel_v1.0.lean`, distributed as a coordinated component in the later machine-backed application packages cited below.

The second is **semantic translation authority**. An adapter or contract must faithfully represent the scientific object, its scope, its evidence states, and the distinctions that matter to the claim. This is the point at which scientifically meaningful content is mapped into the formal burden structure. A formally valid encoding can still be scientifically poor if this mapping is dishonest or mistaken.

The third is **formal authority relative to the encoded record**. Given the states and relations supplied to it, the machine checks whether the stated conclusion follows under the contract. A clean machine certificate therefore establishes a conditional result about the encoded structure. It does not independently establish that the scientific premises were true or that the translation was faithful.

The operative direction can be summarized as:

domain science → adapter / contract → Universal Kernel → machine certificate → scoped scientific claim

The final step returns authority to the scientific anchor rather than allowing the certificate to speak beyond the claim the domain has licensed. The reverse direction is also possible, but only as pressure rather than command: an application may expose a contradiction, missing discrimination, or unabsorbed structural burden and thereby make the universal layer answerable to revision.

Figure 1. Authority Boundary and Formal Flow.

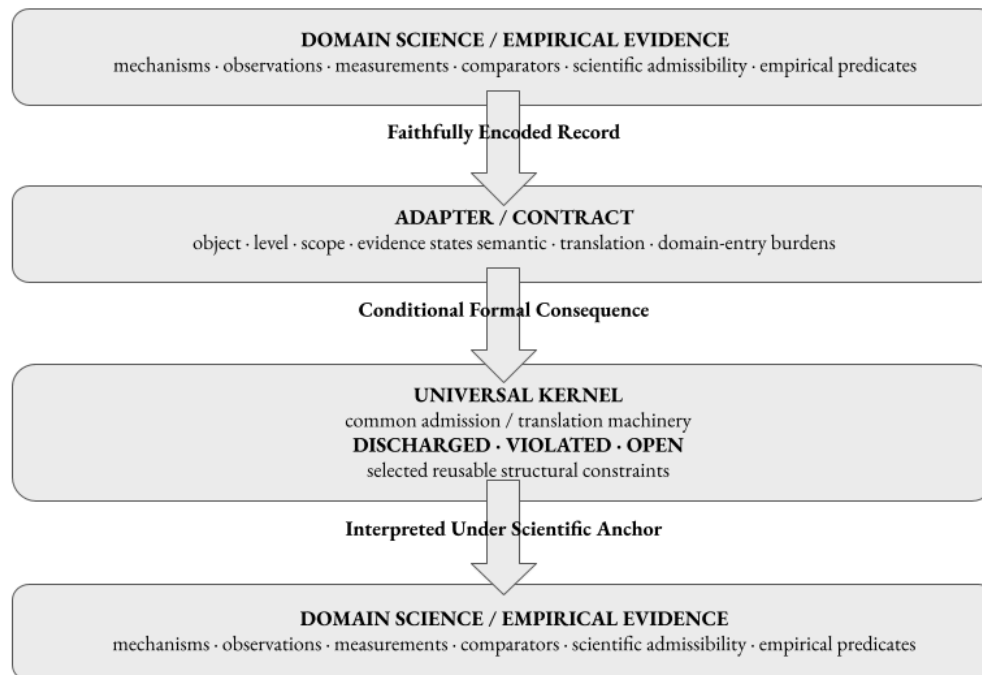


Figure 1 description: Domain science supplies warranted premises and empirical evidence; the adapter or contract translates those commitments at a declared object, level, and scope; and the Universal Kernel checks the conditional consequences of the encoded record. Machine disposition therefore establishes formal status relative to the supplied record, not the empirical truth of the premises or the scientific faithfulness of their translation. Scientific interpretation returns to the domain at the scope actually licensed.

Several reusable structures sit beneath or alongside the common translation interface. They are introduced only where an application needs their particular job. First-break closure distinguishes where a foundational structural route fails rather than treating all breakdowns alike. The three-part object test asks what job a proposed object performs, what its weakest honest form is, and what capacity is lost if that form fails. Post-authorization answerability constrains later continuation to remain sourced through an accounted lineage rather than freely resetting to a pre-update state. Recognition asks whether the same binding content survives lawful local variation without equating that preservation with literal identity. The corpus later shows that these structures are not exercised uniformly; their presence in the kernel does not imply their direct use by every application.

Post-authorization answerability is a useful example of the kernel's non-guarantee posture. Its formal source guard requires a warranted continuation to remain on an explicitly

accounted lineage beginning from the authorized post-state. It excludes an unaccounted reset, but the successor relation itself is intentionally left uninterpreted for the governing domain to supply. It does not determine a metric time, a unique future, or a mechanism of physical succession. Recognition similarly depends on an already authorized binding and an accounted lineage, then asks whether that binding is preserved across a lawful variation of surface or presentation. It does not manufacture the content or decide what the lawful domain variation is.²

This restraint is not incidental. Non-guarantee is upstream of the application results examined in this paper. The architecture does not promise that a posed question will yield an answer, a new object, closure, or even one uniquely privileged downstream question. A lawful inquiry may instead return an earned bounded result, a precisely located blocking burden, an unresolved state, or a negative result where a required burden fails. What non-guarantee forbids is receiving a stronger outcome merely from the existence of the framework. It does not prohibit stronger conclusions once additional scientific and structural burdens have actually been discharged.³

That distinction matters for the meaning of “universal” in this paper. Universal does not mean that every scientific question uses every theorem in the kernel, that every domain shares one ontology, or that domain-specific information should be compressed into one formal vocabulary. It refers to the intended level of the shared machinery: structures admitted there are supposed to perform sufficiently generic, domain-neutral jobs to support lawful reasoning or interoperability across otherwise sovereign domains. A structure does not become universal merely because it recurs locally, appears in machine code, or is useful to several applications. The governing architecture requires separate justification for promotion to the universal layer.

The same rule governs revision. A new scientific complication should be repaired at the lowest authority level capable of carrying it honestly. A local experimental burden remains local if a local contract can represent it without loss. A domain-specific mechanism remains domain-owned even when universal machinery can describe its structural role. Revision pressure reaches the Universal Kernel only when a necessary structural burden cannot be absorbed by existing universal structure, an applicable object-specific layer, or the governing domain. The pressure must identify not merely what is missing, but what job the missing structure performs and what capacity is lost without it.

² From $\neg(\mathcal{V})/\neg$ to Authorization: Deriving the Foundational Hinge Grammar of Structural Flow, v1.5, Zenodo, <https://doi.org/10.5281/zenodo.22147925>; $\neg(\mathcal{V})/\neg$ — A Derivation from Non-Guarantee, Second Edition, v1.5, Zenodo, <https://doi.org/10.5281/zenodo.22736388>.

³ $\neg(\mathcal{V})/\neg$ — A Derivation from Non-Guarantee, Second Edition, v1.5, Zenodo, <https://doi.org/10.5281/zenodo.22736388>.

This creates a deliberately asymmetric architecture. Scientific authority flows into the formal system through warranted premises and translations. The machine checks only their encoded consequences. Scientific conclusions then return to the domain at the scope actually licensed. Structural pressure may travel in the opposite direction, but only after the missing job and its loss have been located strongly enough to make the universal layer answerable.

Figure 2. Structural Pressure and Revision Adjudication.

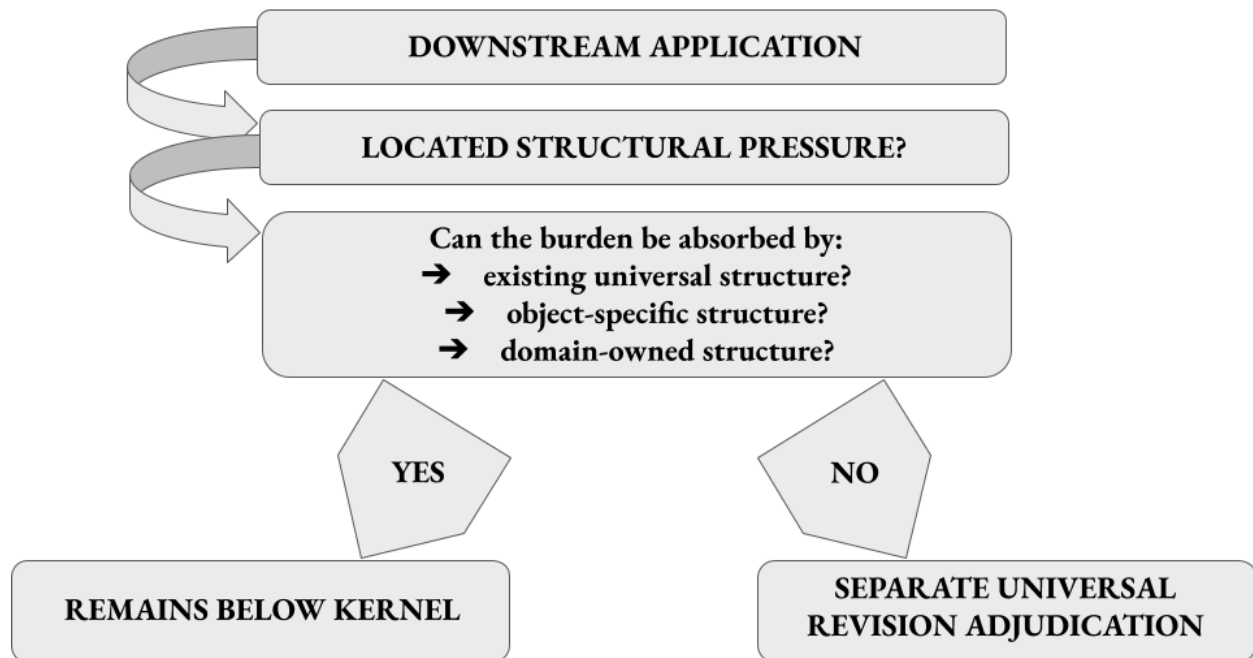


Figure 2 description: Downstream scientific, semantic, or formal work may expose a structural burden that places higher-level machinery under pressure. Such pressure does not itself revise the Universal Kernel. Revision becomes answerable only after the relevant object, level, scope, structural job, and capacity loss have been located and the burden cannot honestly be absorbed by existing universal structure, an object-specific layer, or the governing scientific domain.

The audit that follows tests whether this separation survived contact with heterogeneous scientific problems. If the universal layer remained unchanged only because domain complexity was hidden, the architecture should reveal that weakness in its adapters and coverage. If strong conclusions required unearned outcome assumptions, non-guarantee should fail under application pressure. And if a genuinely reusable structural burden was missing, at least one application should have supplied a reason for the kernel itself to move.

3. Audit Design and Frozen Corpus

The comparative analysis was conducted over a corpus that was frozen before the adversarial cross-application analysis and before the manuscript claims were fixed. The frozen corpus contains eleven source-audited scientific application packages and one separately classified experimental-architecture case in morphogenesis and regeneration. The eleven applications span substantially different scientific questions and modes of inquiry, including failure-regime discrimination, consciousness, quantum gravity, dark energy, the chemistry-to-life boundary, quantum measurement, evolutionary individuality, the thermodynamic arrow, baryogenesis, turbulence, and cosmological-constant mechanisms. The morphogenesis case is retained separately because it reached substantial experimental architecture but was deliberately stopped before protocol freeze or execution.⁴

This corpus is not a random or statistically representative sample of scientific problems. It is better understood as a heterogeneous within-program stress sample: the set of scientific applications that had actually been developed far enough to support source-level comparison before the corpus was frozen. That distinction limits the inference available from the study. The purpose is not to estimate how often the kernel would remain unchanged across science generally. It is to ask a narrower historical and methodological question: across this already heterogeneous set of completed formal applications, what changed locally, what remained shared, and did any application expose a reusable structural requirement that could not be carried by the frozen universal machinery?

Table 1. Frozen Corpus Overview

ID	Scientific domain / question	Disambiguated object under adjudication	Status at corpus freeze	Universal-kernel revision
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⁴ The eleven completed/source-audited public packages in the frozen corpus are: Five Failure Regimes in Dark-Matter Light Probes, coordinated release, Zenodo, <https://doi.org/10.5281/zenodo.23084141>; Consciousness and Experiential Absence, Zenodo, <https://doi.org/10.5281/zenodo.22812662>; When Is Quantum Gravity Required?, Zenodo, <https://doi.org/10.5281/zenodo.22833457>; Where Does the Dark-Energy Contribution Sit?, Zenodo, <https://doi.org/10.5281/zenodo.22845714>; Life from Chemistry, Zenodo, <https://doi.org/10.5281/zenodo.22848429>; Quantum Measurement, Zenodo, <https://doi.org/10.5281/zenodo.22879295>; Evolutionary Individuality as an Empirical Question, Zenodo, <https://doi.org/10.5281/zenodo.22907164>; The Thermodynamic Arrow, Zenodo, <https://doi.org/10.5281/zenodo.22948809>; The Missing Premise in Baryogenesis, Zenodo, <https://doi.org/10.5281/zenodo.22968518>; The Turbulence Identity Problem, Zenodo, <https://doi.org/10.5281/zenodo.23026122>; Cross-Update Stress Tests for Cosmological-Constant Mechanisms, Zenodo, <https://doi.org/10.5281/zenodo.23174925>. The Morphogenesis / Regeneration Scaling case is separate from these eleven completed/source-audited packages and is retained as an ethically terminated architectural stress case.

APP-01	Failure regimes / dark-matter light probes	Five foundational first-break failure classes and their observable discrimination in a light-probe program	Formal and experimental discrimination architecture developed	None earned
APP-02	Consciousness / experiential absence	Evidentiary placement of experiential absence requiring independently admissible positive and negative anchors	Positive anchor present; required negative anchor absent; stronger closure not presently inhabited	None earned
APP-03	Quantum gravity	Conditions under which quantum description is load-bearing, attributable to gravity, and sufficient to defeat an admitted classical gravitational comparison class	Formal experimental-adjudication architecture; empirical execution remains prospective	None earned
APP-04	Dark energy	Physical identification of a baseline-completion contribution within a declared classical metric-GR source/geometry relation	Structural contract closed at declared scope; empirical entry burdens remain open; physical identification not yet adjudicable	None earned
APP-05	Life from chemistry	Organizational continuity through turnover → autonomous maintenance → separate Life translation	Prospective empirical contract; continuity, maintenance, and Life burdens remain separately answerable	None earned
APP-06	Quantum measurement / QCCT	Outcome-conditioned continuation tested through preparation-indistinguishability	Machine-clean contract architecture; physical entry burdens remain	None earned

		and ordered fixed-instrument / process-tensor comparators	open; no empirical QCCT verdict claimed	
APP-07	Evolutionary individuality	A common WS↔SM collective lineage evaluated under separately frozen evolutionary-individuality accounts	Machine-clean target and bounded-resilience branches; proposed experiment not performed	None earned
APP-08	Thermodynamic arrow	Boundary-conditioned Arrow relation under realized-history provenance and declared scope	Theoretical / formal corollary closed at declared scope; physical asymmetry remains domain-owned	None earned
APP-09	Baryogenesis	Whether generation-from-zero is required at a declared history-class scope, conditional on an established zero antecedent	Scope relation formalized; universal antecedent-zero claim remains open	None earned
APP-10	Turbulence	Compound DLBTF $\wedge$ HFN turbulence identity under a frozen scientific scope, with necessity, sufficiency, integrity, redundancy, and residual-identity challenges	Large local formal / experimental architecture; empirical adjudication remains prospective	None earned
APP-11	Cosmological constant	Protection / stability, empirical identification, and numerical realization of a common protected functional across registered updates	Worked cross-update adjudication supports mixed PASS, non-identifying, OPEN, and FAIL endpoints	None earned

TERM-01*	Morphogenesis / regeneration scaling	Patterning competence versus proportional scaling, including donor-associated versus host-associated scale dependence	Substantial causal experiment architecture reached; stopped before vertebrate protocol freeze and execution	None exposed before termination
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*TERM-01 is not counted in $N = 11$. It is retained separately as an ethically terminated architectural stress case.

Freezing the corpus before the adversarial pass was important because further applications remained possible. Without a freeze, a favorable result could be indefinitely strengthened by adding new internally generated cases after the emerging pattern was already visible. The present analysis therefore treats later work as future validation unless the corpus is explicitly reopened and the comparative analysis rerun. The unit of comparison is the frozen application package, not the later trajectory of the broader Structural Flow research program.

The audit was source-based. For each application, the governing scientific artifact or anchor, the associated formal contract or adapter, the relevant Lean source, and the shared Universal Kernel were treated as the evidence-bearing materials. Navigation trackers, maintenance notes, and project summaries were useful for locating files but were not treated as substitutes for the underlying sources. Where a scientific conclusion, formal dependency, scope limit, or kernel-revision claim mattered to the comparison, the audit returned to the package material that actually carried that claim.

This distinction is especially important because the formal system is not treated as an independent source of scientific truth. The governing Structural Flow interface explicitly limits machine authority to the structure actually encoded. It distinguishes structural conformance, failure, unresolved burdens, and current-kernel insufficiency, while reserving scientific meaning and empirical warrant to the relevant domain. A current-kernel insufficiency claim is not licensed merely because an application is difficult or unfamiliar; it requires a located structural burden, an identified structural job and capacity loss, and a demonstration that the burden cannot be absorbed by existing universal structure, an object-specific layer, or the governing domain. The same authority architecture also permits lower-level scientific work to exert upward pressure without allowing local applications to silently rewrite universal structure.

The cross-application coding was designed around that separation of authority. Each application was examined for the scientific problem from which it began, the narrower object or burden architecture that survived disambiguation, the scientific content that remained domain-owned, the parts of the Universal Kernel directly exercised by the adapter, the lawful result states supported by the contract, the treatment of unresolved or failed burdens, the amount and character of local formal structure, and any candidate pressure for universal revision. The final comparative question was deliberately simple: did the application require the shared kernel to change?

The answer to that question was not taken from filenames, version labels, or author recollection. It was checked against the actual formal dependencies and the application's own repair and reopening rules. This matters because an unchanged kernel file can be methodologically uninteresting if all substantive work has merely been hidden elsewhere. The audit therefore treated non-revision as only the beginning of the analysis. It separately examined whether local adapters had enough freedom to absorb arbitrary complexity, whether direct use of the kernel was concentrated in only a small generic interface, whether ostensibly universal requirements had actually been relocated into application-specific code, and whether the formal layer could be kept unchanged only by weakening its scientific role.

The adversarial phase consequently attacked the corpus from several directions. One line of attack asked whether adapter elasticity made kernel stability trivial: if any scientific distinction could simply be encoded locally, then unchanged reuse would say little about the shared machinery. A second asked whether the kernel constrained too little to fail. A third examined direct theorem-family coverage, distinguishing mere inclusion of the same file from actual use of different universal structures. Additional attacks tested whether the scientific objects had been disambiguated before adjudication or merely stipulated for convenience, whether formal and empirical authority had been allowed to leak into one another, whether candidate revision pressures had been assigned to the lowest defensible layer, and whether the absence of independent external replication limited the strength of the comparative result.

A further theoretical pressure test was conducted before claim freeze. It asked what would count as a genuine counterexample to the current kernel rather than a malformed adapter or an out-of-scope philosophical construction. Two routes were distinguished. One would require a faithfully represented contradiction within a structural burden the kernel itself claims to govern, followed by valid formal conformance despite that contradiction. The other would require an independently coherent scientific object that needs a reusable, domain-neutral structural relation the current kernel cannot represent without distortion and that cannot honestly remain local to the object or domain. The purpose of this test was not

to prove that no such case exists, but to make future revision pressure explicit and reproducible.

The morphogenesis program was retained as a separate stress case because its trajectory differs from the eleven application packages in a methodologically useful way. The program reached a retained scientific object, a machine-clean formal contract, a selected vertebrate experimental system, a causal account contrast, a reciprocal transplantation architecture, independent scale coordinates, readout logic, and a prospective outcome matrix. It did not proceed to protocol freeze. The program was explicitly stopped after the ethical and welfare cost of the proposed vertebrate intervention was considered against its prospective scientific gain. The resulting record therefore documents substantial experimental architecture without empirical execution. It is not counted as a completed empirical adjudication.⁵

That distinction also prevents the corpus from being described as a collection of favorable executed outcomes. In the morphogenesis case, the formal contract was judged adequate and the universal kernel remained unchanged, but the scientific program was intentionally parked before execution. The case therefore contributes to the architectural stress test—particularly to the questions of object formation, prospective result routing, and kernel-revision pressure—without being promoted into experimental evidence.

Finally, kernel identity was treated as a reproducibility question distinct from scientific adequacy. The same Universal Kernel file was carried with the application packages, and the archive-recorded checksum provides a public route for testing whether that file moved across releases. The frozen kernel file carried throughout the audited package series is `Structural_Flow_Universal_Kernel_v1.0.lean`. Zenodo records MD5 checksum **4fe6d868650766eba368891ba008ae4d** for the kernel copies used in the audited DOI packages; this archive-recorded MD5 is the available cross-package identity check. The resident audit copy additionally has SHA-256 checksum

b455ec869d062f0fc05d239c5006c843165cd598edbd8c35eae442ff471689a3, reported only as a supplemental local checksum. The complete DOI-to-package MD5 manifest is provided in Appendix F so that readers can independently inspect the archival record.⁶ Checksum identity establishes file identity for the audited release history. It

⁵ Morphogenesis / Regeneration Scaling is an unpublished retained project case. Development reached substantial causal experimental architecture but was terminated before protocol freeze and vertebrate execution. No DOI or public source package is asserted for this case. See Appendix G for the paper's bounded methodological treatment.

⁶ `Structural_Flow_Universal_Kernel_v1.0.lean`. Archive identity check: Zenodo-recorded MD5 **4fe6d868650766eba368891ba008ae4d**. This is the cross-package archival checksum used in the present study. The resident audit copy also has SHA-256 **b455ec869d062f0fc05d239c5006c843165cd598edbd8c35eae442ff471689a3**; that value is reported only as a supplemental checksum of the resident file and is not claimed as Zenodo-recorded across every package. Neither checksum establishes scientific adequacy. Appendix F provides the DOI-to-kernel MD5 manifest.

does not establish that the kernel is complete, sufficient for future applications, or scientifically correct.

The resulting design therefore fixes three boundaries before reporting the comparative results. First, the corpus is heterogeneous but within-program and is not presented as a representative sample of science. Second, the audit distinguishes source-backed scientific and formal claims from project navigation or retrospective summary. Third, kernel non-revision is treated as a historical observation to be explained and attacked, not as a synonym for universality. With those boundaries fixed, the next question is direct: across the frozen corpus, what universal machinery was actually exercised, and did any application earn a change to it?

4. Kernel Non-Revision and Coverage

Across the eleven source-audited application packages, no application demonstrated a reusable, domain-neutral structural requirement that required amendment of the shared Universal Kernel at its declared scope. The separately classified morphogenesis case likewise reached substantial experimental architecture without exposing a kernel repair burden before the program was stopped. This is the most direct comparative result of the audit. It is also deliberately historical and bounded. The result is not that the kernel cannot require revision. It is that none of the frozen applications earned such a revision.

File identity was treated separately from that methodological conclusion. The applications were audited against the same reference kernel, and the archive checks performed during the audit found exact checksum agreement for the package copies directly compared with that reference. The complete archive manifest is provided in Appendix F so that the non-movement claim can be independently checked across the full DOI record. This matters because a claim of formal reuse is stronger when it refers to an unchanged file rather than a sequence of similarly named or incrementally repaired kernels. It remains only an identity claim: an unchanged checksum can show that the file did not move; it cannot show that the file was adequate.

That distinction led to a second audit question. If the same kernel remained unchanged, how much of it was actually being used? Merely packaging one large formal library with every application would provide weak evidence of cross-domain reuse if each adapter touched only one permissive interface. The coverage analysis therefore distinguished direct theorem-family use from inherited semantic structure, application-specific formal machinery, and domain-owned scientific content.

The dominant direct reuse surface was the Universal Translation Contract. In methods terms, this is the common admission and routing interface that keeps the object, level, scope, structural preservation, non-circularity, and identity burdens explicit while allowing object-specific and domain-specific requirements to remain local. Most applications used some substantial portion of this common interface to distinguish structural conformance from required failure, unresolved burden, or downstream domain outcome. Its recurrence is therefore real. It is also the most heavily exercised part of the kernel by a wide margin.

The corpus nevertheless contains direct use beyond that common interface. The Failure Regimes application directly invokes the kernel's first-break closure over the five foundational failure classes. The Life-from-Chemistry application directly invokes the three-part object test that checks whether a candidate preserves its structural job, its weakest honest shape, and the capacity lost when that shape fails, in addition to the general translation contract. That deeper formal use accompanies a scientific architecture that keeps organizational continuity through turnover, autonomous maintenance, and the later Life translation as separate questions rather than allowing one result to manufacture the next. The governing Life/Chemistry architecture explicitly preserves that direction: turnover evidence feeds autonomous-maintenance adjudication, which then feeds a separate Life translation; none of the three steps substitutes for another.

The thermodynamic-arrow application exercises another kernel family directly: post-authorization answerability, or PAA. The relevant formal job is a no-free-reset lineage constraint. Once an admitted history has actually traversed a structurally relevant state, a later continuation cannot simply source itself from an earlier condition while ignoring the intervening realized history. This does not determine a unique future, prohibit lawful return, or supply the physical thermodynamic asymmetry. The physics companion keeps those burdens separate: physics supplies the asymmetric boundary and thermodynamic conditions, while the formal layer checks the conditional consequences, including lawful return, multiple successor outcomes, scope fracture, and explicit failure routes. The application closes against the existing kernel without modifying it.

The cosmological-constant application provides the broadest direct use of deeper kernel machinery in the frozen corpus. Its adapter separately invokes Recognition—the requirement that the same binding content survive across locally variable descriptions—and PAA lineage constraints, while also using the common disposition grammar for a larger successor-experiment ledger. The importance of this case is not simply that more kernel names appear in the source. The adapter contains explicit non-substitution barriers between them. Successful transport of the same binding description does not by itself establish a valid physical-successor history; a valid successor history does not by itself establish protection of

the cosmological baseline; and neither structural result supplies the domain evidence required to establish the physical mechanism. The deeper theorem families therefore perform distinct jobs rather than collapsing into a generic formal “pass.”

This distinction between direct use and scientific dependence is important elsewhere in the corpus. The thermodynamic-arrow argument, for example, requires Recognition and Coherence when local orientations are promoted to one Arrow over a larger scope: the local orientations must remain the same relevant orientation and must jointly hold as one larger-order relation. Those jobs are part of the scientific and semantic architecture of the application, but the released Arrow adapter does not directly instantiate the kernel’s executable Recognition and Coherence theorem modules. They therefore cannot be counted as direct theorem-family coverage merely because the scientific argument refers to the corresponding structural distinctions.

The consciousness application exposes a different version of the same problem. Its generic closure machinery has formal dependencies on the three-part object test and on failure-side discrimination, but the current scientific record lacks the admitted negative anchor required to inhabit the proposed closure packet. The dependency is therefore real while the current scientific state does not exercise the resulting closure theorem. Counting that case as though a completed theorem path had been traversed would conflate formal availability with actual application use.⁷

Once those distinctions are enforced, coverage is visibly uneven. The Universal Translation Contract is recurrent across most of the corpus. FailureClosure, the three-part object test, PAA, and Recognition each receive direct application use in at least one distinct scientific program. By contrast, the audit did not establish direct application-adapter use of several other kernel theorem families, including the executable Coherence module, Equationhood, Density, Ladder, CanonThree, and Duality. This is not a defect discovered by the audit; those structures were not required merely to make the coverage table look balanced. It does, however, set a firm ceiling on what the corpus can demonstrate.

⁷ For the application packages supporting the direct-use distinctions summarized in this section, see Five Failure Regimes in Dark-Matter Light Probes, <https://doi.org/10.5281/zenodo.23084141>; Life from Chemistry, <https://doi.org/10.5281/zenodo.22848429>; The Thermodynamic Arrow, <https://doi.org/10.5281/zenodo.22948809>; Cross-Update Stress Tests for Cosmological-Constant Mechanisms, <https://doi.org/10.5281/zenodo.23174925>; and Consciousness and Experiential Absence, <https://doi.org/10.5281/zenodo.22812662>.

Table 2. Direct Universal-Kernel Theorem-Family Coverage

Kernel surface / theorem family	Direct use established in frozen corpus	Audit interpretation
Universal Translation Contract / common disposition interface	Recurrent across most application packages; dominant shared surface	Broadest exercised interface; applications do not necessarily exercise identical burdens
FailureClosure / first-break closure	APP-01	Distinct direct use beyond the common translation surface
PlusThree / three-part object test	APP-05	Distinct direct use; APP-02 has generic dependency but its current closure route is not inhabited
Post-Authorization Answerability (PAA)	APP-08; APP-11	Direct lineage / no-free-reset work in two different physics applications
Recognition	APP-11	Direct executable use; performs a job distinct from PAA
Failure-side discrimination	No completed direct closure established; generic dependency in APP-02	Formal availability is not counted as completed theorem exercise
Executable Coherence module	No direct application-adapter use established	Scientific / semantic Coherence burdens may appear without exercising the executable theorem module

Equationhood	None established	Not stress-tested by the frozen application corpus
Density	None established	Not stress-tested by the frozen application corpus
Ladder	None established	Not stress-tested by the frozen application corpus
CanonThree	None established as direct application-adapter use	Do not infer coverage from its use in upstream reasoning
Duality	None established	Not stress-tested by the frozen application corpus

Interpretive ceiling: the corpus demonstrates repeated reuse of one common translation / adjudication surface plus several distinct deeper theorem families in a minority of applications. It does **not** demonstrate exhaustive theorem-surface validation.

The resulting claim is therefore narrower than “the whole kernel was validated across eleven domains.” It was not. What the corpus demonstrates is repeated reuse of one common translation and adjudication surface, together with selective direct reuse of several deeper theorem families where particular applications required them. Other parts of the kernel remain weakly tested or untested by this application corpus.

That narrower result is methodologically more useful than an inflated one. Kernel non-revision cannot be explained away simply by saying that none of the applications touched anything beyond a generic wrapper: some did. Nor can repeated packaging of the same file be treated as exhaustive formal validation: much of the kernel remains outside the directly exercised surface. The historical result lies between those two extremes. Heterogeneous applications repeatedly entered through the same shared formal interface; several demanded additional reusable structure already present in the kernel; none demonstrated a missing universal structural job that could not honestly remain at a lower layer.

The remaining question is therefore not whether application complexity existed. It plainly did. The question is where that complexity went. If increasingly elaborate scientific objects, evidence grammars, comparators, failure routes, and experimental programs could all be represented while the universal layer stayed fixed, the next task is to determine whether that stability reflects a meaningful separation of reusable structure from domain-owned science or merely an adapter layer elastic enough to absorb anything.

5. Where the Scientific Complexity Went

The unchanged kernel did not imply simple applications. In several cases the opposite occurred: as the scientific question became more precise, the local formal and experimental architecture became substantially more elaborate. This makes adapter elasticity the strongest immediate challenge to the non-revision result. If every difficult distinction can simply be moved into an application-specific layer, then a stable universal kernel could be achieved by design rather than earned through reuse.

The turbulence program provides the sharpest version of that objection. Its universal dependency is comparatively small, while the local scientific contract is extensive. The turbulence-specific layer must keep separate the physical identities of the flows, their experimental roles, fidelity requirements, transfer-support behavior, history-family behavior, protected holdouts, negative comparators, causal-novelty controls, provenance burdens, anti-replication tests, redundancy checks, and residual-identity challenges. The adapter does not claim to adjudicate those physical facts. It maps scientifically adjudicated states into the shared formal contract; Lean then checks the conditional structure supplied to it. The companion is explicit that a machine-clean result does not establish that a flow is turbulent, that the declared scope is physically correct, or that the relevant experimental burdens were in fact satisfied.

That local complexity is not merely decorative. Different failures have different scientific consequences. A violated constitutive burden can challenge the proposed turbulence identity; an inadequate assay may block adjudication without supporting the opposite result; a post-outcome change in scope or protocol can invalidate the confirmatory route; an established simpler criterion can make the proposed identity redundant; and a surviving independent same-level burden can show that the proposed compound criterion is incomplete. The local contract therefore distinguishes five different routes from a generic formal violation rather than allowing the word “violated” to carry one scientific meaning.⁸

⁸ The Turbulence Identity Problem: A Falsifiable Experimental Criterion, coordinated machine-backed release package, Zenodo, <https://doi.org/10.5281/zenodo.23026122>.

This is precisely why adapter growth cannot be treated as evidence that the universal layer already contains the science. It does not. The turbulence example instead shows what stable formal reuse looks like when much of the scientific burden properly remains local. The common kernel supplies a small disposition and admission grammar; the turbulence package supplies the physical identities, test roles, measurements, comparators, and domain-specific implications needed to use that grammar responsibly.

The important question is therefore not whether local layers become complicated. They should, when the science is complicated. The relevant test is whether the complexity remains local because it performs a genuinely local scientific job, or whether it conceals a reusable structural requirement that the universal layer ought to carry. That distinction recurs throughout the corpus.

In the quantum-gravity application, for example, the broad question “is gravity quantum?” is not sent directly to the machine. The application first separates several independently warranted burdens: whether the tested system has the relevant quantum description, whether that quantum structure is load-bearing for the observed effect, whether the effect is attributable to gravity, and whether an admitted classical gravitational comparison class remains adequate. The experimental architecture then adds a further coverage problem. Closing a finite set of discriminator branches does not establish that the intended classical comparison class has been exhausted. A separate coverage bridge is therefore required, and its physical adequacy remains domain-owned. The machine checks that the bridge cannot simply be omitted when promoting local experimental exclusions to the broader claim.⁹

The same pattern appears in the chemistry-to-life program. The original umbrella question becomes three different scientific jobs: continuity through turnover, autonomous maintenance, and a later Life translation. Evidence that material or organizational continuity survives turnover does not by itself establish autonomous maintenance; autonomous maintenance does not by itself establish Life; and neither result supplies strict identity. The local Life/Chemistry architecture therefore carries assays, control efficacy, endogeneity, integration, regeneration, continuity classes, and other biology- and chemistry-owned burdens while preserving explicit firewalls between the conclusions. None of those burdens becomes universal merely because the application requires them.¹⁰

The cosmological-constant work makes the same move in a different domain. Rather than treating “the cosmological-constant problem” as one indivisible question, the application

⁹ When Is Quantum Gravity Required? A Finite Experimental Adjudication Framework, coordinated machine-backed release package, Zenodo, <https://doi.org/10.5281/zenodo.22833457>.

¹⁰ Life from Chemistry, coordinated machine-backed release package, Zenodo, <https://doi.org/10.5281/zenodo.22848429>.

separates protection or stability, empirical identification, and numerical realization. A mechanism may genuinely protect a relation across perturbative or effective-theory changes without becoming empirically identifiable on the tested observables, and neither result establishes that the mechanism selects the realized numerical value. The application therefore fixes a common protected functional, $\Pi [X; \theta_0]$, requires the same physical job to persist across the tests included in a joint claim, distinguishes legitimate theory-owned evolution from post-result retuning, and separately asks whether a physical successor state is actually discriminating. Here again, the local architecture grows because the scientific claims are not interchangeable.¹¹

Morphogenesis shows the same object-narrowing before empirical execution. The program did not ask simply whether an organism “knows its size.” It separated retained local patterning from proportional scaling and then constructed an experiment capable of asking whether regenerated extent answered more strongly to donor-associated or host-associated scale. Those scale coordinates were to be fixed before the regenerative result was observed. The program reached substantial causal experimental architecture before the vertebrate protocol was deliberately stopped; the draft already preserves its separate status as architectural rather than empirical evidence.¹²

Across these cases, the recurring pattern is not one of adding arbitrary formal thickness after the fact. Broad problem labels are repeatedly decomposed before decisive adjudication into smaller scientific jobs that can fail separately. The resulting local burdens often become numerous because they preserve distinctions the universal layer would be wrong to erase: experimental coverage is not gravity attribution; turnover continuity is not autonomous maintenance; protection is not empirical identification; patterning competence is not proportional scaling; an inadequate turbulence assay is not a necessity falsifier.

Table 3. Representative Object-Disambiguation Cases

Application	Umbrella question	Disambiguated scientific jobs	Critical non-substitution

¹¹ Cross-Update Stress Tests for Cosmological-Constant Mechanisms: A Pre-Specified Framework Across Perturbative, EFT, and Physical-State Changes, coordinated machine-backed release package, Zenodo, <https://doi.org/10.5281/zenodo.23174925>.
¹² Morphogenesis / Regeneration Scaling is an unpublished retained project case. Development reached substantial causal experimental architecture but was terminated before protocol freeze and vertebrate execution. No DOI or public source package is asserted for this case. See Appendix G for the bounded methodological record used in this paper.

Life from Chemistry	When does chemistry become Life?	Organizational continuity through turnover; autonomous maintenance; separate Life translation	Continuity $\neq$ autonomous maintenance $\neq$ Life
Quantum Gravity	Is gravity quantum?	Quantum description; load-bearing quantum structure; gravity attribution; classical comparison-class coverage	Finite discriminator closure $\neq$ exhaustion of the admitted classical comparison class
Cosmological Constant	Does a mechanism solve the cosmological-constant problem?	Protection / stability; empirical identification; numerical realization	Protection $\neq$ identification $\neq$ numerical selection
Morphogenesis / Regeneration Scaling	Does a regenerating system “know” its size?	Patterning competence; proportional scaling; donor-associated versus host-associated scale dependence	Loss of patterning competence $\neq$ negative evidence for the scaling relation
Turbulence	What makes a flow turbulent?	DLBTF $\wedge$ HFN candidate identity; necessity; sufficiency; integrity; redundancy; residual-identity challenge	An inadequate assay or blocked adjudication $\neq$ falsification of the candidate identity

Interpretive ceiling: repeated disambiguation is a corpus-level methodological pattern. The audit does not establish that these decompositions are unique, metaphysically privileged, or independently reproducible.

That pattern provides some counterpressure against the strongest form of the adapter-elasticity objection. The applications are not free to become arbitrarily convenient without consequence. Their scopes are fixed, their constituent burdens can fail, and the local contracts commonly include routes by which the candidate object can be rejected, left unresolved, or reopened. Turbulence is especially instructive because its elaborate local

architecture does not merely protect the proposed DLBTF $\wedge$ HFN identity. The same contract contains sufficiency challenges from negative comparators, necessity challenges from required positives and protected holdouts, redundancy challenges from simpler criteria, and a residual-identity test asking whether an independent same-level identity burden has been omitted. A sufficiently flexible adapter could have been written to protect the candidate from those outcomes; this one was not.

The result nevertheless has a firm ceiling. The corpus does not prove that every local burden has been assigned to the correct authority layer, or that no adapter can hide a missing universal distinction. Nor does repeated object decomposition establish that the resulting scientific objects are unique. Independent investigators may construct rival decompositions, and a future application may reveal a structural job that appears repeatedly across domains and cannot honestly remain local. Such a case would create genuine revision pressure.

What the present corpus supports is narrower. Scientific complexity did not disappear when the kernel remained fixed. It accumulated primarily in the layers that owned the scientific distinctions: domain definitions, evidence requirements, comparators, experimental controls, failure routes, and claim-specific formal contracts. The shared layer remained comparatively austere because these local structures were not promoted merely because they were important.

This gives the non-revision result a more substantive interpretation than simple file stability. Across the audited applications, the architecture repeatedly separated reusable structural requirements from scientific content that changed with the problem. The separation was imperfectly tested and remains open to attack, but it was not costless: local candidates could fail without rewriting the kernel, and local complexity could increase without automatically becoming universal structure.

Figure 3. What Changed, and What Did Not.

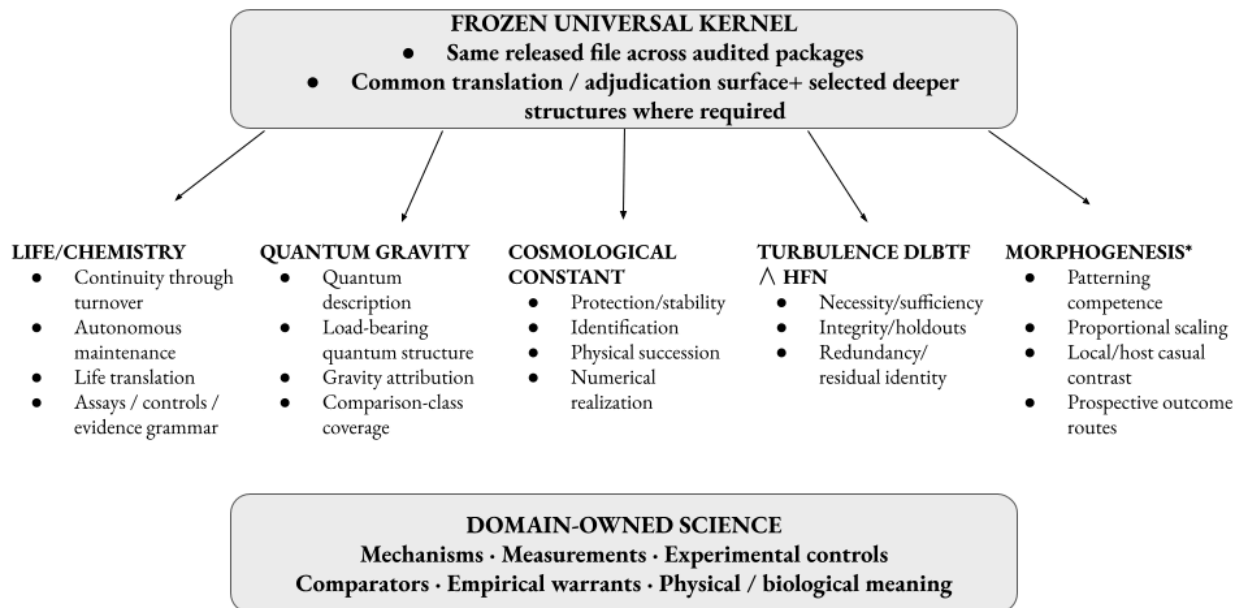


Figure 3 description: Representative applications accumulated substantially different scientific objects, evidentiary burdens, comparator structures, experimental controls, and local formal architecture while retaining the same frozen Universal Kernel. The figure is qualitative: branch size or visual complexity does not measure scientific difficulty, evidentiary strength, or formal coverage. It illustrates the narrower architectural result that substantial local differentiation did not, within the frozen corpus, require automatic promotion of that complexity into universal structure.

The next comparative question is whether that austerity weakened the results. A universal framework that refuses to guarantee outcomes could preserve itself by refusing to decide very much. The corpus therefore has to answer a second challenge: when the applications reached strong bounded conclusions, did they do so only by smuggling outcome selection into local contracts, or could positive, negative, unresolved, and non-identifying results remain lawful parts of the same adjudication architecture?

6. Non-Guarantee and Lawful Non-Answer

The preceding sections leave a second possible explanation for kernel stability. Perhaps the shared layer remained austere because it simply refused to decide very much. A framework that guarantees no particular outcome can always appear robust if every difficult question is

allowed to remain unresolved. The relevant issue is therefore not whether non-guarantee permits openness. It plainly does. The stronger question is whether non-guarantee remained compatible with conclusions strong enough to matter scientifically.

In the Structural Flow architecture, non-guarantee applies not only to physical outcomes but to inquiry itself. A posed question is not guaranteed to yield an answer, a new object, or closure. The lawful result space instead distinguishes an earned bounded answer, a specifically located blocking burden, an unresolved state, and a negative result where a required burden is shown not to hold. Openness is therefore not defined as failed success. It is one possible adjudicative endpoint when the available warrant has reached its limit.

That distinction becomes scientifically substantive only if the different endpoints remain separated when an application produces mixed results. The cosmological-constant program provides the clearest case in the frozen corpus. Its worked example examines original global vacuum-energy sequestering through several different scientific questions rather than asking for one aggregate verdict on the mechanism. The registered matter-sector protection relation survives the tested perturbative and effective-theory challenges. On the local-history QCD and electroweak channels, however, the mechanism is exactly degenerate with the frozen matched-GR comparator and is therefore not identifiable on those channels. The direct whole-history relation remains open because the required complete-history information is not presently available. A separately registered linear-potential completion fails at its stated joint cosmological role. These endpoints coexist in one adjudication without being normalized into a single PASS or FAIL.¹³

The value of that result grammar is easiest to see by considering the alternatives. If exact local degeneracy were reported as empirical failure, a genuine protection result would be lost. If the unresolved whole-history burden were reported as failure, absence of access would become evidence against the mechanism. If failure of one completion were promoted to failure of the parent protection relation, a downstream negative result would erase an independently supported upstream one. The contract explicitly prevents those substitutions. A completion failure remains completion-specific, and an open whole-history burden remains unanswered rather than being redescribed as failed or merely below experimental sensitivity.

This example also shows why non-guarantee should not be confused with refusal to draw strong conclusions. “Not identifiable on these channels” is stronger than “we do not know.” It states that, under the frozen comparator and channel set, the proposed mechanism and

¹³ Cross-Update Stress Tests for Cosmological-Constant Mechanisms: A Pre-Specified Framework Across Perturbative, EFT, and Physical-State Changes, Zenodo, <https://doi.org/10.5281/zenodo.23174925>.

comparator are exactly degenerate. Likewise, failure of the registered completion is a genuine negative conclusion at that completion's declared role. The strength of the inference comes from the scope and burden that have actually been discharged, not from forcing every branch toward one preferred terminal state.

The thermodynamic-arrow application reaches the same methodological point through a different structure. Post-authorization answerability constrains a realized history from freely resetting its provenance, but it does not select a unique future. The formal contract therefore permits accounted return and multiple successor outcomes consistent with the same lineage constraint. It can also return not-yet-adjudicable where a required physical entry burden remains open, no-lawful-translation where an entry or lineage burden fails, and narrower surviving Arrow partitions where a larger claimed scope fractures. The same machine closure thus supports strong conditional structure without becoming a deterministic future generator.¹⁴

The chemistry-to-life program demonstrates the same discipline at a prospective experimental boundary. Its evidence grammar distinguishes discharged, violated, and open burdens, and explicitly states that OPEN is neither FALSE nor FAILED. Organizational continuity through turnover has multiple scientifically meaningful result classes rather than one success state and one generic negative state. Autonomous maintenance can be positive, negative, or indeterminate depending on which burdens have actually been answered. Even a successful autonomous-maintenance result does not automatically become a Life result: the later translation remains separately answerable.

That separation permits procedural closure without pretending that the chemistry is already known. The evidentiary burdens and result grammar can be frozen before an experiment is performed, while the empirical state remains open. A future result can therefore be strong without having been implicitly chosen in advance. If continuity is established but autonomous-maintenance burdens remain unresolved, autonomous maintenance remains unresolved. If an autonomous-maintenance object is eventually admitted but the later Life translation fails, the autonomous-maintenance result remains standing while the Life claim does not. The point of formalization in this setting is not to predict the chemistry but to prevent interpretive rules from moving after the evidence arrives.¹⁵

The Consciousness application supplies an even cleaner stop condition. Its current evidence ledger contains a positive experiential anchor but no admitted negative anchor. Several candidate absence cases remain indeterminate. Because the proposed closure packet requires

¹⁴ The Thermodynamic Arrow: A Boundary-Conditioned Corollary of Realized History and Scope, Zenodo, <https://doi.org/10.5281/zenodo.22948809>.

¹⁵ Life from Chemistry, Zenodo, <https://doi.org/10.5281/zenodo.22848429>.

both sides, the machine can prove that the current ledger is not placement-ready and that the corresponding closure candidate cannot yet be inhabited. It does not convert the missing negative anchor into one, and it does not treat the positive evidence as sufficient to manufacture the absent side of the comparison.¹⁶

That is a non-answer, but not an empty one. It identifies exactly what kind of evidence is missing and why the stronger conclusion cannot yet be licensed. In the terms used by the universal inquiry interface, the work has become more answerable even though it has not reached the requested empirical closure. The distinction matters because an unresolved burden with a known job is methodologically different from an unstructured statement that more research is needed.

Morphogenesis preserves the same boundary prospectively. The contract supports positive scale-conditioned results, bounded negative scale-relation results, and open or indeterminate scale evidence. It also prevents a perturbation that destroys the patterning competence required to assess scaling from being promoted into evidence that the scaling relation itself is absent. If the experiment destroys the capacity needed to ask the question, the corresponding route remains open rather than becoming a false negative.¹⁷

Across these cases, the common pattern is not merely that several result labels are available. The result states preserve different epistemic and scientific jobs. A negative result requires evidence that a required burden fails. An open result preserves an unresolved burden without manufacturing its negation. A non-identification result can be stronger than an ordinary null result because it may establish exact comparator-relative degeneracy. A bounded positive result remains bounded rather than silently inheriting adjacent claims. And multiple lawful successors can remain possible even after a strong structural constraint has been established.

Table 4. Non-Guarantee Outcome Grammar

Lawful result form	Methodological meaning	Representative corpus example

¹⁶ Consciousness and Experiential Absence: An Evidentiary Contract and Staged Experimental Program, Zenodo, <https://doi.org/10.5281/zenodo.22812662>.
¹⁷ Morphogenesis / Regeneration Scaling is an unpublished retained project case. The scaling-discrimination architecture was developed prospectively but the vertebrate protocol was not frozen or executed. No DOI or public source package is asserted for this case. See Appendix G.

Earned bounded positive	Required burdens close at a declared object, level, and scope	Cosmological-constant protection relation passes registered update tests
Negative / violated	Adequate evidence shows that a required burden fails	Registered cosmological-constant completion fails its declared joint role
OPEN / not yet adjudicable	A required burden remains unresolved; no stronger lawful result is available	Consciousness lacks the required admitted negative anchor; Life / QCCT retain open empirical entry burdens
Located blocking burden	The stronger claim cannot close, but the exact missing job and owner are identifiable	Quantum-gravity comparison-class coverage; Consciousness negative-anchor requirement
Non-identifying	The tested evidence cannot distinguish the candidate from the registered comparator at that scope	Cosmological-constant QCD / electroweak local-history channels
Composite / mixed	Different scientific jobs lawfully return different endpoints without mutual rewriting	One cosmological-constant mechanism returns protection PASS, local non-identification, whole-history OPEN, and completion FAIL
Multiple lawful successors	A strong structural constraint holds without selecting one unique continuation	Thermodynamic Arrow permits more than one successor compatible with the same provenance constraint
Scope-bounded conditional	A stronger burden follows only where its antecedent has independently been established at the same scope	Baryogenesis: $\text{ZeroAntecedent}(S) \rightarrow \text{GenerationFromZeroBurden}(S)$

Key distinction: OPEN is not FALSE. A negative result requires a warranted violation; absence of closure does not manufacture one.

This is the principal sense in which the corpus moves from non-guarantee toward answerability. Non-guarantee does not supply the answer. It constrains what may be claimed before the answer is earned. As an application accumulates warranted structure, the lawful result space narrows: some candidate interpretations become unavailable, some burdens become precisely located, some negative results become licensed, and some positive conclusions close at a declared scope. The absence of a universal outcome guarantee therefore need not imply weak adjudication.

The frozen corpus supports a bounded version of that claim. Across the audited applications, no strong scoped closure or empirically actionable adjudication required the Universal Kernel to introduce a premise whose job was to preselect the scientific outcome. Stronger conclusions were instead obtained by adding warranted domain premises, predeclared comparators, failure conditions, and application-specific burdens below the universal layer. Where those premises did not close, the framework retained the weaker lawful state rather than supplying the missing answer universally.

That observation does not establish that non-guarantee is sufficient for every scientific problem. A future application may expose a class of inquiry in which useful answerability cannot be achieved without some additional reusable structural requirement. Nor does the existence of multiple result states guarantee that a particular adapter has assigned those states correctly. Those questions belong to the adversarial limits of the architecture.

What the present corpus does show is that non-guarantee was compatible with considerably more than agnosticism. It coexisted with positive closure, scoped falsification, exact non-identification, explicit empirical blockers, unresolved questions with located burdens, and formal constraints that preserved more than one lawful future. The next task is therefore to turn the analysis against itself: which of these apparent successes remain vulnerable to weak-kernel triviality, semantic mispopulation, circular object formation, limited theorem coverage, or the absence of independent external replication?

7. Adversarial Stress Test and Surviving Limitations

The preceding results survive only in a bounded form. The adversarial pass did not locate a frozen-corpus application that required amendment of the Universal Kernel, nor did it locate a case in which a faithfully represented required kernel-owned failure was nevertheless licensed as structural conformance. That does not dispose of the strongest objections to the architecture. Several remain live, and each limits a different inference that might otherwise be drawn from kernel stability.

The first is adapter elasticity. Section 5 showed why local formal complexity is not by itself evidence against a reusable universal layer: many of the distinctions carried locally are scientifically specific and would be distorted by universalization. The turbulence package is the clearest example. Its adapter explicitly maps domain adjudications rather than independently deciding the physics, and the meanings of open, violated, or discharged depend on the scientific burden at which those states occur. That architecture supports the narrower claim that a stable interface can coexist with substantial domain-owned complexity.

It does not establish that every burden currently left local belongs there. A sufficiently elastic adapter can protect an underspecified kernel if new distinctions are always permitted to accumulate downstream. The relevant unresolved question is therefore whether some future local structure will recur strongly enough, and with enough structural necessity, that continuing to call it application-specific becomes artificial. The present corpus did not expose such a burden. That is an observed result, not a proof that none exists.

Closely related is the weak-kernel or triviality objection. If the shared layer constrained almost nothing, reuse would be easy. The corpus narrows that objection but does not eliminate it. Direct application use was not limited to a single generic wrapper: several applications exercised deeper kernel structures, and some of those structures block inferences that would otherwise be tempting. The quantum-gravity contract, for example, includes a formal countermodel showing that successful closure of six experimental discriminator blocks does not by itself establish exhaustion of the admitted classical gravitational comparison class. The independently warranted coverage bridge remains necessary before that promotion can occur. The kernel and associated formal contracts therefore do constrain at least some nontrivial inferential moves.

At the same time, Section 4 established that theorem-family coverage is uneven. Much of the kernel has not been directly exercised by the frozen application adapters. The corpus therefore cannot support a claim that every part of the kernel has survived heterogeneous scientific stress. What has been tested repeatedly is the common translation and adjudication surface, together with selected deeper structures. Kernel stability and exhaustive kernel validation remain different claims.

A more fundamental limitation concerns the boundary between formal soundness and semantic faithfulness. The machine can check only the record it receives. The governing interface states this directly: the machine workbench may audit only what it encodes. If a required burden is faithfully represented as unresolved or violated, the formal architecture can block the stronger inference. But the kernel does not independently establish that the

scientist chose the correct burden, populated its evidence honestly, identified the correct scope, or supplied a scientifically faithful interpretation of the encoded state.

This is not a peripheral implementation issue. It marks the boundary of machine authority. A formally conforming record can be scientifically misleading if its domain predicates were badly chosen or falsely populated. Conversely, a scientifically valid result can fail to enter the formal system if the translation is defective. The appropriate claim is therefore conditional: given a scientifically valid and faithfully populated record, the machine checks the consequences encoded by the contract. Machine conformance alone is not evidence that the record is scientifically faithful.

The application packages themselves preserve this distinction. The turbulence companion states that Lean can check whether supplied states satisfy encoded definitions or whether a formal blocker is constructible, but cannot establish whether the flow, intervention, measurement, scope, or physical provenance is actually correct. The quantum-gravity formalization similarly leaves comparison-class coverage as an explicit domain-owned proposition: Lean checks that the burden cannot be omitted from the promotion step, not that the physical coverage claim is true. The formal/empirical gap therefore survives the audit by design. The architecture makes that gap inspectable; it does not eliminate it.¹⁸

The same issue creates a residual circularity threat around object disambiguation. Across the corpus, broad scientific questions repeatedly became narrower objects or burden architectures before adjudication. This appears to be an important recurring feature of the method. But the applications were developed within the same originating research program that developed the Structural Flow machinery. The audit can show that the resulting objects were given failure conditions, that they could remain open or fail, and that post-result mutation was constrained. It cannot show that Structural Flow uniquely discovers the scientifically correct decomposition.

A skeptic could therefore argue that the method succeeds because the problem is reformulated into an object the framework can handle. The existing corpus supplies some counterpressure: the resulting objects are often allowed to generate inconvenient outcomes, including non-identification, unresolved empirical burdens, failed completions, insufficient comparison-class coverage, and absent negative anchors. The Consciousness contract, for example, retains a positive anchor while refusing closure because the required negative anchor is absent; its not-yet-adjudicable state is explicitly neither rejection nor empirical

¹⁸ For explicit machine-authority limits in application packages, see The Turbulence Identity Problem, <https://doi.org/10.5281/zenodo.23026122>, and When Is Quantum Gravity Required?, <https://doi.org/10.5281/zenodo.22833457>. Both coordinated packages separate domain-owned physical premises from the conditional formal inference checked against `Structural_Flow_Universal_Kernel_v1.0.lean`.

closure. Such cases are difficult to reconcile with a procedure that simply defines its way to positive answers.¹⁹

They still do not establish decomposition uniqueness. Independent investigators may identify different scientifically defensible objects, different burden partitions, or different stopping points. The important future test is not whether an outsider uses the same vocabulary, but whether independently produced decompositions survive comparable scope, failure, non-circularity, and evidence burdens—and whether incompatible decompositions can be discriminated rather than merely coexisting under equally permissive translations.

That external test has not yet occurred. The frozen corpus is a within-program heterogeneous stress sample. Independent specialists have not yet taken the frozen kernel, selected a new scientific problem, and independently carried the full object-to-contract process far enough to constitute a separate validation population. Independent reproduction of the existing object decompositions is likewise not yet available. This is best described as an unperformed validation test, not as demonstrated dependence on the author. The work has therefore earned a cross-domain within-program result, not a population-level claim about how the method transfers between researchers.

The empirical status of the corpus imposes a separate ceiling. Several applications close theoretical relations, formal contracts, or experimentally answerable architectures rather than reporting newly executed experiments. Quantum gravity remains an empirical adjudication proposal; Life-from-Chemistry preserves a prospective contract while current evidence remains open; Morphogenesis stopped before vertebrate execution; Consciousness retains an unresolved empirical anchor burden. Machine-clean formalization and strong experimental specification can make these questions substantially more answerable, but they are not substitutes for the experiments or observations required to settle their domain-owned premises.

Chronology also requires care. Several applications contain pre-outcome freezes, fixed comparators, prospectively declared failure rules, or result taxonomies designed to resist retrospective reinterpretation. Those controls matter. They should not be inflated into a claim that every exploratory derivation in the broader research program was independently preregistered before its scientific target became visible. The paper audits the frozen packages and their internal information-order protections; it does not reconstruct an independently witnessed preregistration history for every step that led to them.

¹⁹ Consciousness and Experiential Absence: An Evidentiary Contract and Staged Experimental Program, Zenodo, <https://doi.org/10.5281/zenodo.22812662>.

A related archival limitation was resolved for publication by assembling the complete DOI-to-kernel manifest in Appendix F. File identity was checked against the reference kernel and the archive-recorded package checksums used in the audit. This makes the historical non-movement claim independently inspectable at the file-identity level. It does not strengthen the scientific claim: checksum agreement establishes identity of the archived kernel file, not its adequacy, completeness, or scientific correctness.

Finally, the absence of revision in the frozen corpus cannot be converted into a forecast of permanent non-revision. The governing architecture itself prohibits such promotion. Local recurrence does not establish universality merely by occurring more than once, and machine residency does not elevate a structure into universal semantic authority. A later application may reveal a structural job that the present kernel lacks. If that job is genuinely necessary, reusable, domain-neutral, and cannot honestly be carried by existing universal machinery or a lower authority layer, then the historical result reported here becomes the starting point for revision rather than an argument against it.

The adversarial pass therefore leaves the central findings standing but smaller than their strongest possible interpretation. The corpus supports stable reuse of a frozen formal interface under substantial local scientific variation. It supports strong bounded answerability without universal outcome preselection. It supports the claim that no audited application earned kernel amendment at its declared scope. It does not establish kernel completeness, unique object decomposition, automatic semantic faithfulness, exhaustive theorem-family coverage, independent reproducibility, or empirical confirmation of programs that have not been executed.

Table 5. Adversarial Threats and Surviving Limitations

Threat	What would make it serious	Counterpressure found in corpus	Status after audit
Adapter elasticity	Any scientific distinction can be hidden locally, leaving the universal layer vacuous	Local contracts contain genuine failure, OPEN, redundancy, non-identification, and reopening routes; some applications exercise deeper kernel structures	Survives; narrowed, not eliminated

Weak-kernel / triviality objection	Shared machinery constrains almost nothing beyond bookkeeping	FailureClosure, PlusThree, PAA, and Recognition perform distinct direct jobs in selected applications	Narrowed; not eliminated
Semantic mispopulation	Machine conformance is treated as proof that the encoded scientific record is faithful	Architecture explicitly separates scientific premises, semantic translation, and formal consequence	Major surviving limitation
Object-disambiguation circularity	Problems are reformulated only into objects the method can “solve”	Objects can fail, remain OPEN, return non-identification, or block stronger closure	Survives; uniqueness not established
Uneven theorem coverage	Same file is mistaken for broad testing of every theorem family	Audit separates direct executable use from generic dependency or semantic reliance	Survives as firm claim ceiling
Independent external replication	Cross-domain stability depends on originating research program rather than transferable method	Public frozen packages and explicit failure criteria make independent attack possible	Not yet tested
Empirical execution gap	Machine-clean or experimental architecture is mistaken for performed science	Paper explicitly separates formal closure, empirical entry, and executed experiment	Survives by design; empirical programs remain domain-owned

Chronology / preregistration inflation	Prospective controls are described as though every exploratory step was independently preregistered	Individual packages contain freezes, fixed comparators, and prospective failure rules	Claim must remain package-specific
Archive-identity incompleteness	“Same kernel” cannot be independently checked across releases	Zenodo records the same kernel MD5 where the kernel is present	Resolved for publication: Appendix F provides the full DOI → kernel → MD5 manifest
Pairwise → global Coherence overreach	Pairwise certificates are silently promoted to arbitrary n-ary / global coherence	Released executable scope is explicitly pairwise	Live future pressure candidate, not current falsification

Overall adversarial disposition: no current-kernel revision was earned by the frozen corpus, but adapter elasticity, semantic faithfulness, decomposition uniqueness, uneven theorem coverage, and independent transfer remain live limitations.

Those remaining vulnerabilities suggest a more useful next step than another internally generated application. The architecture should be exposed to attacks capable of changing its status. The relevant question is no longer merely whether a skeptic disagrees with an application. It is what a skeptic would have to demonstrate to show that the frozen kernel itself falsely licenses a result or excludes a coherent scientific object it ought to be able to represent.

8. External Falsification and Revision Pressure

The limitations identified above become scientifically useful only if they can be converted into attacks that could change the status of the framework. Mere disagreement with an application is not enough. Nor is the discovery of an unresolved scientific question, an awkward local contract, or a domain-specific mechanism absent from the universal layer. The

relevant challenge is narrower: identify a failure at the level where the kernel itself claims authority, or identify a structural job that the kernel ought to carry but presently cannot.

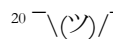
The framework therefore admits two principal routes to genuine kernel pressure.

The first is **false conformance**. Consider an independently admissible scientific object translated at a declared level and scope. Suppose a required structural burden within the kernel's own authority is genuinely violated or unresolved, and suppose that fact is faithfully represented in the record supplied to the formal system. If the kernel nevertheless licenses the corresponding structural conformance, then the failure reaches the universal machinery. That would not be an empirical disagreement downstream of the kernel. It would be a counterexample to the kernel's own adjudication logic.

The qualification "faithfully represented" is essential. Section 7 established that the machine cannot infer whether a scientist has hidden a contradiction, mislabeled evidence, omitted a necessary predicate, or supplied a dishonest witness. A malformed scientific record can produce a misleading formal result without showing that the formal implication itself is defective. A false-conformance test must therefore preserve the contradiction inside a burden the kernel actually claims to inspect. Only then does a positive machine result place the kernel rather than the adapter under direct pressure.

The universal translation interface already states the relevant expected behavior. A required violated burden blocks lawful translation; a required unresolved burden blocks structural conformance; and structural conformance is available only after the required universal, object-specific, and domain-entry burdens have been discharged or lawfully routed. The same interface reserves a distinct insufficiency state for cases in which an exact required structural burden has been located but cannot be absorbed by existing universal, object-specific, or domain-owned structure. A faithful case that violates these relations would therefore constitute a direct framework-level failure.²⁰

The second route is **coherent-object exclusion**. Here the kernel need not falsely license anything. Instead, an independently defensible scientific object would expose a structural relation required for lawful adjudication that the present kernel cannot represent without distortion. For that result to pressure the universal layer, the missing relation would have to perform a genuinely reusable and domain-neutral job. It would also have to resist lawful placement in the object-specific contract or governing scientific domain. Most new scientific detail will fail this test, appropriately: a new assay, physical mechanism, biological criterion,

²⁰  — A Derivation from Non-Guarantee, Second Edition, v1.5, Zenodo, <https://doi.org/10.5281/zenodo.22736388>; see also the Universal Translation Contract implemented in `Structural_Flow_Universal_Kernel_v1.0.lean` within the coordinated application packages cited in this paper.

comparator family, or measurement rule does not become universal merely because an application needs it.

The governing revision rule therefore asks for more than novelty. Upward pressure must locate the object, structural level, scope, exact missing burden or failed discrimination, the job performed by that burden, the capacity lost or blocked without it, the nearest existing structure that might absorb it, and the lowest level at which the problem cannot honestly be repaired. If an application can make that showing and the missing job is genuinely reusable across domains, the current kernel becomes answerable to revision.

These routes also clarify what does **not** constitute a kernel falsification. An adapter with invented evidence does not defeat the kernel. A contradiction hidden outside the encoded record does not show false conformance. A scientifically important burden that is honestly domain-owned does not establish missing universal structure. A new local distinction does not become universal because it appears in more than one application. A failed experiment can overturn a domain identification without touching the formal architecture that made the experiment answerable. And a proof-engineering failure, by itself, does not show that a new structural capacity is required.

Figure 4. Two Routes to Genuine Kernel Pressure.

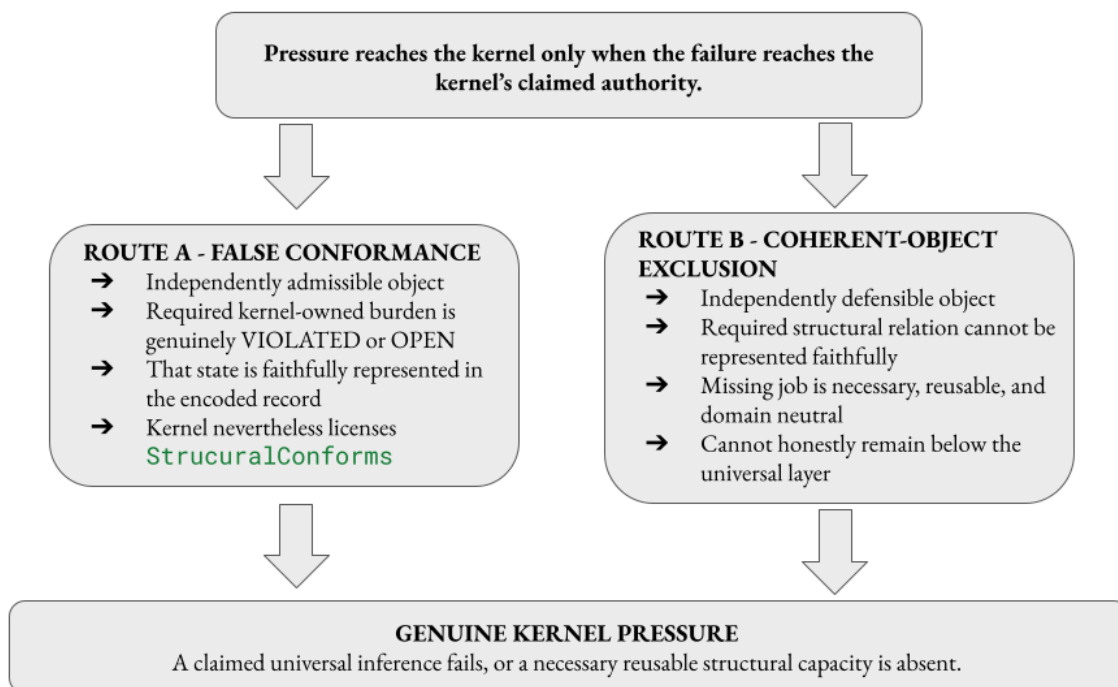


Figure 4 description: Kernel-level pressure can arise in two distinct ways. False conformance occurs when a required kernel-owned failure or unresolved burden is faithfully represented in the encoded record yet the kernel nevertheless licenses structural conformance. Coherent-object exclusion occurs when an independently defensible scientific object requires a necessary, reusable, domain-neutral structural capacity that the current kernel cannot faithfully carry and that cannot honestly remain below the universal layer. Either result would make the current kernel answerable to revision; neither implies that revision has already been adjudicated or approved.

Scope violations are equally important. One live pressure candidate identified during the audit concerns Coherence. The released executable Coherence machinery explicitly certifies a pairwise, binary minimum: its formal world contains two local pieces and one candidate whole, and the release declares the current executable scope as pairwise. Pairwise compatibility and pairwise whole-hold cannot therefore be silently promoted into a claim of arbitrary global coherence.²¹

This matters because pairwise compatibility does not in general guarantee compatibility of an entire multi-component family. A future application could contain three or more locally admissible structures whose pairwise relations all succeed while their simultaneous realization is globally frustrated. Such a case would not falsify the current pairwise theorem: the theorem does not claim n-ary coverage. It would, however, become genuine revision pressure if the global relation performed a reusable structural job that could not be represented through lawful composition of the existing pairwise machinery or honestly retained at a lower layer.

The pairwise→n-ary boundary is therefore a useful example of how the framework is intended to fail well. The present scope is explicit enough that an out-of-scope case cannot be counted as a hidden contradiction. At the same time, the scope declaration prevents the existing theorem from being used to dismiss a scientifically necessary broader relation. If such a relation is eventually demonstrated, the appropriate result is not that the earlier pairwise theorem was false, but that the universal interface requires a justified extension.

External application provides the strongest available test of both routes. An independent researcher need not accept Structural Flow terminology or reproduce the original derivational path. A more demanding test would begin with the frozen kernel and a scientific problem not selected during construction of the present corpus. The researcher would

²¹ Structural_Flow_Universal_Kernel_v1.0.lean, Coherence module, current executable scope declared pairwise / binary. The archive-recorded cross-package identity check used in this study is MD5 4fe6d868650766eba368891ba008ae4d. For the foundational public release lineage, see “(C)” — A Derivation from Non-Guarantee, Second Edition, Zenodo, <https://doi.org/10.5281/zenodo.22736388>.

independently locate the scientific object, declare its level and scope, identify the relevant domain-owned burdens, and attempt the translation without modifying the universal machinery in advance.

Several outcomes would be informative. The application might close or become empirically answerable using the frozen kernel while accumulating substantial local structure, extending the historical stability result beyond the originating program. It might expose a legitimate missing universal capacity and thereby earn kernel revision. It might produce a faithfully populated record for which the machine licenses an invalid structural conclusion. It might yield a rival object decomposition that survives the same scope, failure, and non-circularity pressures while supporting a materially different adjudication. Or it might fail at some more ordinary scientific or formal boundary. None of these outcomes needs to be converted into confirmation of the framework. Each would locate a different answerable burden.

This is why independent application is more valuable at this stage than simply increasing the number of internally authored examples. The frozen corpus already contains enough heterogeneity to expose substantial variation in domain object, local architecture, evidence state, theorem-family use, and lawful result structure. Another internally generated case could add breadth, but it would not directly answer the strongest remaining validation question: whether the same separation of universal and local burden survives when object formation and scientific translation are performed by someone who did not build the architecture.

The falsification program can therefore be compressed into a deliberately austere instruction:

Test it. Try to break it. Locate the job and the loss.

A successful challenge need not overthrow every application. It need only show where a claimed universal inference fails, or where a necessary reusable structural capacity is absent. Conversely, failure to locate such a defect in a new independent application would extend the observed stability of the kernel without proving its completeness.

The methodological objective is thus not to make the kernel unrevisable. It is to make revision pressure locatable. A universal formal architecture earns little from being difficult to criticize. It earns more if a critic can state exactly what is missing, at what scope, which capacity is lost without it, and why the burden cannot honestly remain elsewhere. In that sense, disagreement becomes most useful when it can be converted into an executable research program.

9. Discussion

The strongest observation in the frozen corpus is also the simplest: the applications changed substantially; the shared kernel did not. That observation is methodologically interesting only if both halves are preserved. If the applications had remained formally and scientifically similar, unchanged reuse would say little. If the kernel had remained fixed only because it carried no consequential constraints, non-revision would be trivial. The preceding analysis instead found substantial variation in scientific object, evidence structure, local formal architecture, admissible outcomes, experimental posture, and direct theorem-family use, while no audited application earned amendment of the shared Universal Kernel at its declared scope.

The result should not be read as evidence that the kernel is finished. Non-revision means only that the frozen corpus did not expose a reusable, domain-neutral structural job that could not be represented by existing universal machinery or lawfully retained at a lower authority level. This is a historical result about a declared corpus. A future application may expose exactly such a burden. If it does, revision would not contradict the methodological purpose of the architecture. It would demonstrate that revision pressure had finally been located.

This distinction changes the meaning of stability. A reusable formal kernel need not be valuable because it never changes. It may instead be valuable because changes can be assigned to the level that owns them. In the applications studied here, scientific complexity often increased dramatically without becoming universal structure. Quantum-gravity coverage remained a physics burden. Autonomous maintenance and continuity remained Life/Chemistry burdens. Turbulence required a dense local identity and violation-routing architecture. Cosmological-constant mechanisms required distinctions among protection, identification, physical succession, and numerical realization. None of those distinctions became universal merely because they were difficult or scientifically important.

Reusability in this sense is therefore not domain erasure. The corpus does not support an interpretation in which heterogeneous sciences are reduced to one vocabulary or one ontology. The universal layer supplies a comparatively austere interoperability surface; the scientific content remains thick, local, and domain-owned. The resulting architecture resembles a contract boundary more than a universal scientific model. Different domains may supply radically different premises and evidence while still being answerable to some of the same structural questions: what object is under review, at what level and scope, what has been carried across a translation, what remains unresolved, and which inference is or is not licensed by the record.

That separation also helps explain why adapter complexity is not automatically evidence against the kernel. A local adapter becomes suspicious when it absorbs a distinction that performs a reusable universal job, or when it protects a preferred result from failure. It is not suspicious merely because the local science is complicated. Indeed, demanding that the kernel directly encode every mechanism, assay, comparator, measurement convention, or domain-specific failure condition would destroy the very authority separation the framework is intended to preserve. The relevant question is not how much local code exists, but whether the local burden can honestly remain local without loss of a reusable structural capacity.

The corpus does not fully settle that question. Adapter elasticity remains one of the principal surviving threats. The turbulence case makes the issue especially visible: a large and scientifically specific local architecture can coexist with a comparatively small universal interface. The present audit found counterpressure against the strongest triviality interpretation because the local contracts contain genuine routes to failure, openness, redundancy, non-identification, or reopening, and because several applications directly exercise deeper kernel structures beyond the common translation surface. Even so, a future cross-domain recurrence may reveal that some burden currently treated as local has become too structurally generic to remain there. The architecture must remain answerable to that possibility.

Non-guarantee provides a second interpretation of the observed stability. The Universal Kernel was not constructed by collecting all of the scientific features that appeared across the application domains. Its governing posture is more restrictive: stronger outcomes, identities, mechanisms, and closure relations are not admitted merely because an inquiry would benefit from them. What survives at the universal level is therefore structure intended to remain usable without supplying the scientific answer in advance.

The frozen corpus suggests that this refusal need not produce weak inquiry. Non-guarantee coexisted with bounded positive closure, explicit negative results, exact non-identification, located empirical blockers, unresolved states, and multiple lawful successors. The relevant methodological contrast is therefore not between guarantee and ignorance. It is between an outcome received for free and an outcome earned by additional burdens. A framework can remain non-guaranteeing at the universal level while supporting strong conclusions at a narrower scientific scope once the required premises and evidence have been supplied.

That is the principal sense in which the paper's title moves from non-guarantee to answerability. Answerability is not a promise that every question terminates in a positive answer. It is the progressive localization of what the question depends on, what would count

as support or failure, which stronger claims remain unavailable, and what evidence could change the present state. An OPEN result with a located burden may therefore represent more methodological progress than a nominal answer whose supporting assumptions remain hidden.

The recurring object-disambiguation pattern may be related to this result, but the present corpus does not establish the causal relation. Broad scientific questions were repeatedly decomposed into narrower objects and separately answerable jobs before formal adjudication. That process often prevented neighboring claims from rising and falling together: continuity was separated from autonomous maintenance, protection from identification, finite experimental discrimination from comparison-class exhaustion, and local patterning from proportional scaling. One plausible hypothesis is that kernel stability depends less on the raw complexity of a domain than on whether the scientific object can be stated at a level and scope that permits domain-specific burdens to remain where they belong.

That hypothesis is not yet a result. The same research program developed both the Structural Flow method and the application decompositions. The audit did not compare those decompositions against blinded independent alternatives, and it did not establish that the selected objects are unique. A different investigator may decompose the same scientific problem differently while remaining scientifically coherent. The important future question is whether those rival objects can be made answerable to comparable scope, failure, and non-circularity burdens, and whether the current kernel can represent them without distortion.

This unresolved validation burden is now more important than adding another internally generated application. The existing corpus is already heterogeneous enough to have exposed meaningful differences in local formal size, evidence mode, direct kernel use, experimental maturity, and lawful result structure. Additional internal cases may still reveal genuine revision pressure, but their marginal value for validating transfer is lower. The more consequential test is whether an independent specialist can take the frozen public machinery, choose a problem not selected during construction of the corpus, construct the scientific translation independently, and either make the architecture work or locate where it fails.

Peer review can provide one form of adversarial pressure, particularly against claim scope, scientific interpretation, formal correctness, and overlooked alternatives. It is not equivalent to independent application. A reviewer may identify a defect without attempting to use the framework, while a successful review process may occur without anyone independently

reproducing the object formation or translation procedure. The stronger transfer test therefore remains external use.

The same authority separation that limits the present claims makes such testing possible. The machine need not be granted authority over the science for its failures to be informative. An independent researcher can challenge the empirical premises, the semantic translation, or the formal implication separately. A scientifically poor premise should fail at the scientific layer. A distorted translation should fail at the semantic interface. A faithfully populated record that produces an invalid formal license should fail at the kernel. This separation does not guarantee that disagreements will be easy to resolve, but it makes clearer what kind of disagreement is actually occurring.

The frozen corpus is therefore informative without being population-level evidence. Its value lies in the combination of heterogeneity and explicit claim ceilings. It contains enough variation to show that local scientific architecture can grow substantially while a common formal interface remains stable; enough outcome diversity to show that non-guarantee is compatible with more than unresolved inquiry; and enough adversarial structure to identify several ways the interpretation could still fail. It does not establish that the same pattern will survive independent use.

The next validation step is accordingly straightforward in principle, even if difficult in practice: expose the frozen kernel and its application record to investigators who did not construct them and allow the method to become answerable to their scientific objects. If the architecture is missing a reusable structural job, an external application should be able to locate it. If the kernel falsely licenses a contradiction within its own authority, that failure should be reproducible. If rival object decompositions survive the same pressures, the method must account for them.

The present result is therefore best treated neither as proof of universality nor as a preliminary curiosity. It is a bounded record of cross-domain stability under a specific methodological constraint: scientific answers were not installed at the universal level in advance. Across the frozen corpus, substantial local complexity accumulated, strong scoped results remained possible, unresolved burdens remained visible, and no application earned a repair to the shared kernel. Whether that pattern survives independent application is now an open empirical and methodological question.

10. Conclusion

This study began with a methodological constraint: do not give the scientific inquiry its outcome for free. The resulting Universal Kernel was designed to preserve reusable structural burdens while leaving mechanisms, measurements, evidence, comparators, and empirical truth with the scientific domains that own them. The question examined here was whether such an austere shared layer could remain useful when applied to substantially different scientific problems.

Across the frozen corpus of eleven completed source-audited application packages, plus one separately classified terminated experimental-architecture case, the local scientific and formal structures changed substantially. Objects were narrowed, evidence grammars changed, adapters grew or remained small, different kernel structures were exercised, and lawful outcomes ranged from positive and negative conclusions to non-identification, unresolved burdens, composite results, and multiple admissible successors. No audited application, at its declared scope, demonstrated a reusable domain-neutral structural requirement that required amendment of the shared Universal Kernel.

That is a historical result, not a proof of universality. The corpus does not establish that the kernel is complete, minimal, optimal, unique, permanently stable, or sufficient for every future scientific problem. Direct theorem-family coverage is uneven. Some applications contain extensive local architecture. Semantic faithfulness remains a human and domain-scientific burden rather than something machine conformance can guarantee. Independent researchers have not yet reproduced the application process or tested the frozen kernel on newly selected problems.

Those limitations do not erase the narrower methodological result. Scientific complexity did not have to disappear for a common formal interface to remain stable. Much of that complexity could remain where it belonged: in domain definitions, evidentiary requirements, experimental controls, comparators, measurement semantics, and claim-specific contracts. The machine could then check conditional structural consequences without becoming the source of the scientific premises it received.

Non-guarantee likewise did not reduce the corpus to agnosticism. Strong bounded conclusions remained possible when their supporting burdens were discharged. Where those burdens were not discharged, the architecture could preserve a weaker result without manufacturing the missing premise. OPEN did not have to become FALSE. Failure at one scientific job did not have to erase success at another. A valid lineage constraint did not have to select a unique future. Answerability increased as the location of the remaining burden became more precise.

The principal unresolved test is now external. The next important result need not be another successful application. An independent application may instead identify a missing universal structural capacity, demonstrate invalid conformance from a faithfully populated record, produce a rival object decomposition that survives the same pressures, or expose some other precise failure. Any of those outcomes would be informative.

The value of a reusable formal kernel is therefore not that it prevents future revision. Its value is that revision pressure can be made explicit.

If the current kernel is missing a reusable structural job, a future application should be able to say where.

If it falsely licenses a contradiction within its own authority, that failure should be reproducible.

If the scientific burden belongs elsewhere, the architecture should permit it to remain there without silently promoting it into universal structure.

The frozen corpus records a simpler fact:

**the applications changed;
the shared kernel did not.**

For now, that is the result. The next burden belongs to the test.

Appendices

Appendix A. Frozen Corpus Manifest

This appendix fixes the application corpus used in the comparative audit. The corpus contains eleven public, source-audited application packages, designated APP-01 through APP-11. Morphogenesis / Regeneration Scaling is retained separately as TERM-01 because it reached substantial causal experimental architecture but was deliberately terminated before vertebrate protocol freeze or execution. TERM-01 is not included in $N = 11$.

The unit of comparison is the frozen public application package and its declared scientific scope. “Completed/source-audited” refers to completion of the auditable package used in this study; it does not imply that every proposed experiment was executed or that every empirical burden was discharged.

ID	Frozen public artifact	Scientific domain / question	Audited status	Version-specific DOI	Public package posture	Empirical posture at corpus freeze	Universal -kernel revision
APP-01	Five Failure Regimes in Dark-Matter Light Probes: An Open-Source Pre-Result Demonstration Protocol	Failure-regime discrimination / dark-matter-sensitive light probes	Completed/source-audited public package	10.5281/zenodo.23084141	Four-layer: scientific anchor + specialist companion + domain adapter/contract + frozen Universal Kernel	Pre-result experimental protocol. No completed simulation campaign, numerical calibration, target unblinding, real-data analysis, or empirical confirmation is claimed.	None earned at declared scope
APP-02	Consciousness and Experiential Absence: An Evidentiary Contract and Staged Experimental Program	Consciousness / experiential absence	Completed/source-audited public package	10.5281/zenodo.22812662	Four-layer: scientific anchor + machine companion + scoped contract/adaptor + frozen Universal Kernel	Positive experiential anchor admitted; required negative anchor absent. Current placement remains NotYetAdjudicable .	None earned at declared scope

APP-03	When Is Quantum Gravity Required? A Finite Experimental Adjudication Framework	Quantum gravity	Completed/source-audited public package	10.5281/zenodo.22833457	Four-layer: scientific anchor + machine companion + scoped formal contract + frozen Universal Kernel	Experimental-adjudication framework. No claim that quantum gravity has already been empirically required at a tested scope.	None earned at declared scope
APP-04	Where Does the Dark-Energy Contribution Sit? A Geometry–Growth Test of Baseline, Source-Sector, and Modified-Coupling Interpretations	Dark energy	Completed/source-audited public package	10.5281/zenodo.22845714	Four-layer: scientific anchor + human-readable contract companion + scoped Lean contract + frozen Universal Kernel	Experiment derived but not executed. Empirical entry burdens remain OPEN; physical identification is not yet adjudicable.	None earned at declared scope
APP-05	Life from Chemistry: An Abiogenesis Experiment for Autonomous Maintenance	Chemistry-to-Life boundary	Completed/source-audited public package	10.5281/zenodo.22848429	Four-layer: scientific anchor + contract/adaptor companion + scoped Life / origin-of-Life contract and adaptor + frozen Universal Kernel	Prospective empirical contract. OCTT and AMC states remain indeterminate / OPEN; AMC → Life translation is not yet adjudicable.	None earned at declared scope
APP-06	Quantum Measurement: Separating the Physical Burdens and Testing Outcome-Conditioned Continuation	Quantum measurement	Completed/source-audited public package	10.5281/zenodo.22879295	Four-layer: scientific anchor + QCCT audit/execution companion + consolidated contracts/adapters + frozen Universal Kernel	Pre-result. Seam G and QCCT-I remain not yet adjudicable; no empirical QCCT verdict is claimed.	None earned at declared scope

APP-07	Evolutionary Individuality as an Empirical Question: A Proposed Experimental Test in <i>Pseudomonas</i>	Evolutionary individuality	Completed/source-audited public package	10.5281/zenodo.22907164	Four-layer: scientific anchor + adapter/contract companion + scoped domain adapter/contract + frozen Universal Kernel	Proposed experiment not performed. Empirical burdens remain specialist-owned and prospectively adjudicable.	None earned at declared scope
APP-08	The Thermodynamic Arrow: A Boundary-Conditioned Corollary of Realized History and Scope	Thermodynamic arrow	Completed/source-audited public package	10.5281/zenodo.22948809	Four-layer: scientific anchor + machine companion + scoped contract/adapter + frozen Universal Kernel	Theoretical/formal corollary closed at declared scope; physical asymmetry and boundary conditions remain domain-owned.	None earned at declared scope
APP-09	The Missing Premise in Baryogenesis: The Scope of the Zero Antecedent	Baryogenesis	Completed/source-audited public package	10.5281/zenodo.22968518	Three-layer: scientific anchor + scoped formal contract/adapter + frozen Universal Kernel	Conditional scope relation formalized. Universal antecedent-zero across the full admissible history class remains OPEN; no actual cosmic history is selected.	None earned at declared scope
APP-10	The Turbulence Identity Problem: A Falsifiable Experimental Criterion	Turbulence	Completed/source-audited public package	10.5281/zenodo.23026122	Four-layer: scientific anchor + specialist companion + turbulence-specific adapter/contract bundle + frozen Universal Kernel	Candidate empirical identity and prospective adjudication protocol. Machine-clean execution is not empirical turbulence confirmation.	None earned at declared scope

APP-11	Cross-Update Stress Tests for Cosmological-Constant Mechanisms: A Pre-Specified Framework Across Perturbative, EFT, and Physical-State Changes	Cosmological-constant mechanisms	Completed/source-audited public package	10.5281/zenodo.23174925	Four-layer: scientific anchor + contract/adaptor companion + CCP stability-routing adaptor + frozen Universal Kernel	Worked cross-update adjudication. Different registered scientific jobs lawfully return protection PASS, local non-identification, whole-history OPEN, and completion FAIL without mutual rewriting.	None earned at declared scope
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TERM-01 — Unpublished Ethically Terminated Architectural Stress Case

Morphogenesis / Regeneration Scaling is retained separately from the eleven public application packages. The program reached substantial causal experimental architecture, including separation of retained patterning from proportional scaling and a prospective causal test of donor-associated versus host-associated scale dependence. Development was terminated before vertebrate protocol freeze and empirical execution after ethical and welfare review.

TERM-01 has **no public DOI asserted in this study**, contributes **no empirical result**, and is **not counted in N = 11**. Its relevance is methodological: substantial object disambiguation, result routing, and causal experimental architecture were reached without exposing a Universal-Kernel repair burden before termination.

The corpus used throughout this paper is therefore:

11 completed/source-audited public application packages + 1 separately classified ethically terminated architectural stress case.

Appendix B. Direct Kernel Theorem-Family Matrix

This appendix expands the theorem-family coverage summary in Table 2. It records **application-adaptor use**, not mere presence of a theorem family in the distributed Universal Kernel.

The classification is intentionally conservative:

DIRECT — executable application-adapter use was established in the source audit.

GENERIC / DEPENDENCY — a formal dependency exists, but the relevant completed theorem route is not presently inhabited by the application state.

SEMANTIC ONLY — the scientific or explanatory architecture uses the corresponding distinction, but direct executable use of the kernel theorem family was not established.

NONE ESTABLISHED — no direct use of that theorem family was established in the audit.

Kernel-file inclusion is not counted as DIRECT. Ordinary Structural Flow reasoning used during derivation is likewise not counted as application-adapter theorem coverage.

B.1 Translation, failure, object, and lineage surfaces

Application	Universal Translation Contract	FailureClosure	PlusThree	FailureDiscrimination	PAA
APP-01 — Failure Regimes	NONE ESTABLISHED	DIRECT	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED
APP-02 — Consciousness	DIRECT	NONE ESTABLISHED	GENERIC / DEPENDENCY	GENERIC / DEPENDENCY	NONE ESTABLISHED
APP-03 — Quantum Gravity	DIRECT	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED
APP-04 — Dark Energy	DIRECT	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED
APP-05 — Life from Chemistry	DIRECT	NONE ESTABLISHED	DIRECT	NONE ESTABLISHED	NONE ESTABLISHED

APP-03 — Quantum Gravity	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED
APP-04 — Dark Energy	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED
APP-05 — Life from Chemistry	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED
APP-06 — Quantum Measurement	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED
APP-07 — Evolutionary Individuality	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED
APP-08 — Thermodynamic Arrow	SEMANTIC ONLY	SEMANTIC ONLY	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED
APP-09 — Baryogenesis	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED
APP-10 — Turbulence	NONE ESTABLISHED	SEMANTIC ONLY	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED
APP-11 — Cosmological Constant	DIRECT	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED	NONE ESTABLISHED

B.3 Application-specific audit notes

APP-01 — Failure Regimes. Direct use of **FailureClosure** was established through the five foundational first-break classes. No additional direct theorem-family use is inferred merely from the presence of the full Universal Kernel in the package.

APP-02 — Consciousness. The application uses the common translation/adjudication interface for its present OPEN empirical-entry state. Its generic closure machinery also depends on PlusThree and failure-side discrimination, but the required negative experiential anchor is absent. The corresponding closure route is therefore not presently inhabited, and neither dependency is counted as DIRECT.

APP-03 — Quantum Gravity. The formal contract directly uses the **UniversalTranslationContract** interface and its generic disposition machinery. The finite discriminator architecture, comparison-class coverage bridge, and ontology firewall remain QG-specific formal and scientific structure rather than evidence of additional universal theorem-family use.

APP-04 — Dark Energy. The scoped baseline-completion contract directly instantiates the **UniversalTranslationContract**. Its object-specific structural burdens and dark-energy-specific empirical entry burdens remain local to the application.

APP-05 — Life from Chemistry. The application directly uses both the common translation interface and the PlusThree object test. PlusThree supplies an executable job/weakest-shape/capacity-loss burden for the scoped Life object; OCTT and AMC remain domain-specific scientific architectures rather than new universal theorem families.

APP-06 — Quantum Measurement. The consolidated measurement contracts/adapters use the common Universal-Kernel translation and disposition machinery. Seam G and QCCT-I remain distinct application modules; their shared packaging does not establish additional theorem-family coverage.

APP-07 — Evolutionary Individuality. The domain adapter directly uses the **UniversalTranslationContract** to separate structural conformance from downstream account-specific biological outcomes. Clarke, Bourrat, and life-cycle-closure criteria remain domain-owned structures.

APP-08 — Thermodynamic Arrow. PAA is directly exercised as a no-free-reset lineage constraint. Recognition and Coherence perform genuine jobs in the scientific and semantic architecture when local orientations are promoted toward one Arrow over a larger scope, but the released Arrow adapter does not directly instantiate the executable Recognition or Coherence theorem modules. They are therefore classified as SEMANTIC ONLY.

APP-09 — Baryogenesis. The scoped formal contract uses the common translation/adjudication machinery to preserve the conditional relation between an established zero antecedent and a generation-from-zero burden. The history-class and

antecedent predicates remain physics-owned; no deeper direct theorem-family use was established.

APP-10 — Turbulence. The application directly uses the `UniversalTranslationContract` for its scoped identity/adjudication interface. Direct executable Coherence was considered and rejected because the public pairwise Coherence minimum was stronger than the turbulence translation had earned. Coherence is therefore retained semantically rather than counted as direct theorem use.

APP-11 — Cosmological Constant. The CCP adapter uses the common adjudication surface and directly exercises both Recognition and PAA. Recognition carries description-preservation work; PAA carries successor-lineage / no-free-reset work. Those jobs remain distinct, and neither supplies the physics of the mechanism or successor state.

B.4 Coverage ceiling

The matrix shows a strongly nonuniform coverage profile. The Universal Translation Contract is the dominant shared executable surface. Direct use beyond that interface is concentrated in a small number of applications: FailureClosure in APP-01, PlusThree in APP-05, PAA in APP-08 and APP-11, and Recognition in APP-11.

No direct application-adapter use was established in the frozen corpus for the executable Coherence module, Equationhood, Density, Ladder, CanonThree, or Duality. Scientific or semantic reliance on a similarly named distinction does not alter that classification.

Accordingly, the corpus demonstrates repeated use of a common translation and adjudication interface together with selective direct exercise of several deeper theorem families. It does **not** demonstrate exhaustive validation of the Universal Kernel's theorem surface.

Appendix C. Full Object-Disambiguation Matrix

The comparative audit repeatedly found that broad scientific problem labels were not sent directly to formal adjudication. Each application first fixed a narrower scientific object, burden, or relation at a declared level and scope. The amount of decomposition varied by application. The matrices below therefore record the distinctions actually required by each program rather than imposing one uniform decomposition template across the corpus.

C.1 Scientific object, scope, and critical discrimination

Application	Umbrella scientific problem	Narrowed object / burden architecture	Critical non-substitution	Declared level and scope
APP-01 — Failure Regimes	Can failure-regime structure yield falsifiable dark-matter light-probe predictions?	The irreducible first structural break of an independently warranted dark-matter-facing physical route , classified as Dissolution, Shear, Drift, Echo, Lock, or PASS before the held-out probe result is examined.	First-break class $\neq$ image morphology. Route failure $\neq$ disappearance of the whole astrophysical object. Source consequence $\neq$ propagated light observable.	Route level within an independently warranted dark-matter physical structure, under a declared physical participation burden and comparator.
APP-02 — Consciousness	What would warrant a claim of experiential absence?	C3R: current experiential standing of a bounded consciousness-capable bearer at one fixed passage, adjudicated through independently admissible POSITIVE, NEGATIVE, or INDETERMINATE experiential anchors.	Current experience $\neq$ standing capacity for consciousness $\neq$ conscious content $\neq$ ordinary carried content $\neq$ behavior or physiology $\neq$ memory $\neq$ retrospective report. Absence of report does not supply absence of experience.	Bounded bearer / fixed-passage level. The object is deliberately narrower than a general theory or mechanism of consciousness.
APP-03 — Quantum Gravity	When does nature require a quantum-gravity description?	A scoped quantum-gravity requirement in which quantum description is admitted, quantum structure is load-bearing, gravity attribution is established, and the admitted classical gravitational comparison class fails.	Failure of one classical model $\neq$ exhaustion of the admitted classical class. Closure of finite discriminator blocks $\neq$ parent classical exclusion without a separate coverage warrant. A scoped requirement $\neq$ selection of a microscopic QG ontology.	One exact frozen physical scope, pinned by declared burden, level, and scope under one auditable experimental family.
APP-04 — Dark Energy	Where does the dark-energy-sensitive contribution sit?	Physical identification of the dark-energy-sensitive burden with baseline geometry-side completion B inside an already-qualified classical metric-GR source/geometry relation.	Structural availability of the B seat $\neq$ empirical identification of dark energy with that seat. Baseline completion $\neq$ source-sector contribution $\neq$ modified coupling. Physical seat $\neq$ microscopic origin or ultimate ontology.	Role/relation level of classical metric-GR cosmology, over the declared observational and theoretical scope of the geometry–growth test.
APP-05 — Life from Chemistry	When does chemistry become Life?	A three-stage architecture: organizational continuity through turnover (OCTT) $\rightarrow$ autonomous maintenance (AMC) $\rightarrow$ separate Life translation.	Continuity $\neq$ autonomous maintenance $\neq$ Life. External resource dependence $\neq$ external organizational substitution. Turnover continuity $\neq$ strict identity. Activity, synthesis, copying, or persistence alone $\neq$ Life.	Declared chemical system and environment, with explicit level, scope, horizon, boundary, process set, and control set; Life translation then occurs at the declared living level.
APP-06 — Quantum Measurement	What physical burdens remain inside the quantum-measurement problem, and can outcome-conditioned continuation be tested?	Separate Seam G / Born and QCCT-I objects, with QCCT testing preparation-indistinguishable inputs against fixed standard-instrument / process-continuation comparators.	Calibration $\neq$ Born weighting $\neq$ outcome standing $\neq$ continuation. Preparation-equivalence failure or side-channel detection blocks the target witness; it does not become evidence for the opposite continuation claim.	Frozen experimental preparation / instrument / successor-process scope. Each module retains its own physical entry conditions and conclusion ceiling.

APP-07 — Evolutionary Individuality	When does a collective become an evolutionary individual?	The same evolved WS↔SM collective lineage subjected to a post-evolution scaffold challenge and evaluated separately under higher-level selection, endogenization, and life-cycle-closure accounts.	Higher-level selection capacity ≠ endogenous carrying of individuating organization ≠ closed collective life cycle. Failure under one account does not erase surviving structure recognized by another.	Collective-lineage level in an experimentally evolved <i>Pseudomonas fluorescens</i> system, under one declared scaffold-relaxation intervention.
APP-08 — Thermodynamic Arrow	What grounds a thermodynamic Arrow, and how far does it extend?	A boundary-conditioned Arrow relation carried by realized-history provenance and closed only over the scope at which relevant local thermodynamic orientations are recognized as the same orientation and jointly support one Arrow.	PAA provenance ≠ thermodynamic entropy or an Arrow-generating mechanism. Local Arrow ≠ unrestricted global Arrow. Failure of one larger-scope Arrow does not erase independently viable narrower Arrows.	Realized physical history at a declared thermodynamic / cosmological scope, which may be local, subsystem-level, branch-relative, or broader only where the corresponding scope burdens close.
APP-09 — Baryogenesis	Does the observed matter–antimatter asymmetry require generation from an antecedent zero state?	A scoped conditional burden: ObservedNonzero ∧ ZeroEstablished(S) → GenerationFromZeroBurden(S) , with promotion from a model family to the full admissible-history class treated as a separate physical burden.	Late nonzero baryon asymmetry ≠ established zero antecedent. Zero in a model family ≠ zero across all admissible histories. A possible nonzero antecedent history ≠ the actual history of the universe.	History-class / antecedent-boundary level at declared physical scope S, including the stronger candidate scope H _{adm} only if that promotion is independently warranted.
APP-10 — Turbulence	What makes a flow turbulent?	A candidate compound identity DLBTF ∧ HFN within a frozen first-closure scope Σ_C , challenged through distinct necessity, sufficiency, integrity, redundancy, and residual-identity tests.	Inadequate assay ≠ falsification. Failure of one test role ≠ automatic failure of every identity burden. Machine-clean contract execution ≠ empirical turbulence identification. A simpler sufficient identity may make the compound proposal redundant without making the data meaningless.	Physical flow-identity level within the exact declared first-closure scope Σ_C and frozen test-case roles.
APP-11 — Cosmological Constant	Does a proposed mechanism solve “the cosmological-constant problem”?	At least three separately adjudicated jobs: protection / stability , empirical identification , and numerical realization , with cross-update tests tied to one common protected physical functional $\Pi[X; \theta_0]$.	Protection ≠ empirical identification ≠ numerical selection. Successful description transport ≠ valid physical-successor lineage. Failure of a proposed completion does not erase an independently supported parent protection result.	Mechanism- and claim-relative level across registered perturbative (R _i), EFT/matching (R ₂), and physical-successor (S) updates that concern the same protected physical job.

C.2 Present posture, surviving neighbors, and reopening

Application	What can fail while neighboring structure survives?	Present corpus posture	Principal reopening / status-changing condition
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APP-01 — Failure Regimes	A registered route can fail while the dark-matter structure remains physically present and gravitationally effective; distinct routes through one object may receive different first-break classifications.	Pre-result formal and experimental discrimination architecture developed; no completed dark-matter empirical result claimed.	Reopen if a sixth irreducible route-level first-break class survives the protocol's mixture, incompleteness, equivalence, and reducibility tests; if route admission becomes circular; or if domain evidence defeats the claimed physical discrimination.
APP-02 — Consciousness	Positive experiential evidence can survive while no admissible negative anchor exists. Behavioral, physiological, memory, or report evidence may change without thereby deciding experiential absence.	One POSITIVE anchor admitted; no NEGATIVE anchor admitted; empirical placement and stronger closure remain NotYetAdjudicable .	Status changes if a NEGATIVE anchor is independently admitted under the frozen evidentiary standard, or if the C3R object, admission contract, or passage-specific discrimination is itself defeated.
APP-03 — Quantum Gravity	One classical candidate may fail while another faithful classical representation survives; six-block experimental closure may hold while comparison-class coverage remains inadequate.	Structural adjudication architecture closed; proposed experiment not performed; empirical QG requirement remains open; microscopic ontology unresolved.	Reopen if QDA, QBA, or GBA is defeated; a faithful classical representation survives; comparison-class coverage is shown inadequate; or the frozen experimental / nuisance / comparison architecture no longer supports the intended promotion.
APP-04 — Dark Energy	The structural B-seat relation may remain available while empirical identification is still OPEN; a downstream physical interpretation can fail without erasing the already-qualified source/geometry relation.	Structural / contract-shape burdens closed at declared scope; dark-energy-specific empirical entry burdens remain OPEN; identification remains NotYetAdjudicable .	Reopen rather than reinterpret if evidence requires changing the physical job of B, the comparison seats, the declared scope, or the meaning of an entry burden; a required empirical violation can instead defeat the proposed identification at its stated scope.
APP-05 — Life from Chemistry	Turnover continuity may be established while AMC remains unresolved or negative; AMC may later be admitted while the Life translation still fails or remains OPEN.	Prospective empirical contract; OCTT, AMC, and Life states remain separately answerable rather than collapsed into one present verdict.	Reopen the relevant layer if system / environment / level / scope / horizon / boundary or assay architecture changes materially, or if evidence defeats a substantive AMC or Life-translation burden. Failure downstream does not automatically erase upstream continuity.
APP-06 — Quantum Measurement	A preparation-side-channel problem can block QCCT without defeating the general measurement question; one comparator or generalized foil	Machine-clean architecture; domain-owned probability and QCCT empirical entry burdens	Reopen locally if preparation indistinguishability fails, a side channel survives, a comparator is inadequate, a formula or foil is defective, or the frozen instrument / process

	can fail while other modules remain intact.	remain OPEN; no empirical QCCT verdict claimed.	assumptions change. Local repair need not reopen unrelated modules.
APP-07 — Evolutionary Individuality	Endogenization may fail while higher-level selection capacity or life-cycle closure survives; loss of one account-specific property need not erase every form of collective organization.	Target lineage and bounded-result architecture machine-clean; proposed scaffold-challenge experiment not performed.	Reopen if the biological lineage is not held fixed, the intervention fails to isolate the scaffold relation, relevant account criteria materially change, or evidence shows the compared criteria were not independently adjudicable on the common target.
APP-08 — Thermodynamic Arrow	One larger-scope Arrow can fail while narrower coherent Arrow partitions survive; provenance constraints can remain valid while a claimed common orientation fails to close globally.	Theoretical / formal corollary closed at declared scope; physical asymmetry and boundary conditions remain domain-owned.	Reopen if the asymmetric boundary, realized descendant history, thermodynamic asymmetry, PAA lineage, Recognition, or scope-level Coherence fails, or if a qualifying same-scope no-extra-driver counterexample is established.
APP-09 — Baryogenesis	A baryogenesis mechanism may generate the observed asymmetry without establishing that its antecedent was zero; a nonzero antecedent may be physically possible without being the universe's actual history.	Full admissible-history zero antecedent is OPEN; a fully admitted nonzero-B counterexample at the same scope is also not established.	The stronger scope closes if zero is established exhaustively across H_adm; it is defeated if one fully admissible history has genuinely nonzero net B at the relevant antecedent boundary. Either result remains distinct from selecting the actual cosmic history.
APP-10 — Turbulence	Individual constitutive, necessity, sufficiency, integrity, or redundancy burdens can fail independently; a blocked assay can leave the candidate unresolved rather than falsified.	Large local formal / experimental architecture; empirical identity adjudication remains prospective.	Reopen or reject the candidate if a required constitutive burden is violated, a protected holdout defeats necessity, negative comparators defeat sufficiency, a simpler adequate identity makes the proposal redundant, or an independent same-level residual identity burden survives. Post-result scope mutation also reopens the claim.
APP-11 — Cosmological Constant	Protection can PASS while local channels are non-identifying, a whole-history burden remains OPEN, and a separately registered completion FAILS. None of those results automatically rewrites the others.	Worked cross-update case supports a mixed result: protection PASS, local non-identification, whole-history OPEN, and completion FAIL at their respective registered jobs.	Reopen the affected claim if the registered claim profile, update class, common protected physical job, provenance, comparator warrant, theory data, or supporting physical evidence changes. A downstream completion failure alone does not reopen an independently established parent protection result.

C.3 TERM-01 — Morphogenesis / Regeneration Scaling

TERM-01 is retained separately because it is not part of the eleven completed public application packages.

Its umbrella question—whether a regenerating system “knows” its size—was narrowed to **scale-conditioned morphogenesis**: whether a morphogenetic system may preserve scale-relative pattern or growth as the spatial extent relevant to that process changes.

The critical discrimination was:

patterning survives; proportional scaling fails.

That discrimination prevents loss of general patterning competence from being redescribed as negative evidence for proportional scaling. The experimental architecture further separated donor-associated from host-associated scale dependence, fixed local and host scale coordinates prospectively, and developed a reciprocal transplantation design with nested signaling and final-growth readouts.

The program reached substantial causal experimental architecture but stopped before vertebrate protocol freeze and execution. No empirical result is claimed. No public DOI or source package is asserted for TERM-01, and it is not counted in $N = 11$.

C.4 Interpretive ceiling

Across the frozen corpus, broad scientific problem labels repeatedly became narrower objects, burden partitions, or scoped relations before decisive formal adjudication. The resulting decompositions preserved distinctions that could fail independently and frequently prevented success or failure at one scientific job from silently propagating into neighboring claims.

This matrix records a recurring within-program disambiguation pattern. It does not establish uniqueness or independent reproducibility of the resulting objects.

Appendix D. Outcome and Non-Guarantee Matrix

The application corpus does not use one universal scientific PASS / FAIL vocabulary. The shared formal layer supplies a small disposition grammar, but each application retains the result states required by its own scientific object. A supportive result in one application may therefore be a route-level PASS, an admitted experiential POSITIVE, a continuity class,

standard-instrument closure, a scoped Arrow corollary, or protection of a registered functional. Likewise, an adverse result may be a foundational first break, an admitted NEGATIVE, a violated required burden, failure of a candidate identity, or failure of a separately registered completion.

The matrices below preserve those application-native meanings before comparing their broader methodological roles. In particular, **OPEN, INDETERMINATE, and NOT YET ADJUDICABLE are not treated as negative results.** A negative result requires affirmative warrant that the relevant burden fails.

D.1 Application-native positive, adverse, and unresolved result forms

Application	Bounded positive / supportive result available	Adverse / negative result available	OPEN / indeterminate / blocker result available
APP-01 — Failure Regimes	PASS for a registered physical route that completes its tested burden without one of the five foundational first breaks; successful pre-registered source/probe discrimination may support the declared demonstration claim.	Dissolution, Shear, Drift, Echo, or Lock where the corresponding first break is independently warranted. A genuinely unrepresented irreducible class can challenge the five-class closure.	Inadequate route admission, unresolved mixture or purity, insufficient measurement support, or incomplete evidence can block a stronger route classification. The protocol retains the strongest state fully supported rather than promoting incomplete evidence.
APP-02 — Consciousness	POSITIVE experiential standing where the admission contract warrants current experience at the pinned bearer and passage.	NEGATIVE where a dedicated negative route, adequate temporal anchoring, independence, and required memory-confound controls warrant absence of experiential standing.	INDETERMINATE for unresolved conflict, inadequate anchoring, candidate dependence, capacity-only evidence, or insufficient negative evidence. At the universal interface, the currently missing NEGATIVE anchor yields NotYetAdjudicable .
APP-03 — Quantum Gravity	A scoped quantum-gravity requirement becomes live where QDA, QBA, and GBA are discharged and admitted classical gravitational representability, GCR, is violated.	A surviving faithful classical gravitational representation blocks the claimed QG requirement at that scope; failure of another required positive gate likewise	Any required parent gate or classical-representability question may remain OPEN. Insufficient comparison-class coverage blocks promotion from finite experimental exclusion to the parent classical-exclusion claim.

		blocks the final promotion.	
APP-04 — Dark Energy	Structural Conforms for the proposed baseline-completion identification where all required empirical entry burdens are discharged at the declared scope.	No Lawful Translation for the proposed identification where a required entry burden is adequately shown to be violated.	Not Yet Adjudicable where no required burden is violated but one or more required dark-energy entry burdens remain OPEN.
APP-05 — Life from Chemistry	OCTT can return one of several warranted continuity classes; AMC can return POSITIVE where all admission and substantive burdens are discharged; an admitted AMC object can subsequently enter a separate Life translation.	AMC can return NEGATIVE where pins and assay adequacy are discharged and at least one substantive AMC burden is violated. A later Life translation can independently fail.	OCTT may remain INDETERMINATE ; AMC may remain INDETERMINATE ; individual burdens may remain OPEN . Missing or inadequate assays do not become evidence that autonomous maintenance is absent.
APP-06 — Quantum Measurement	Possible bounded results include preparation-equivalence closure, standard-instrument closure , characterization of a non-Lüders ordinary instrument, process-tensor closure, and, only after registered alternatives fail, a residual showing insufficiency of the registered ordinary comparator at the tested scope.	A named model or foil may be constrained or rejected; a feasibility or entry burden may fail locally; an experimentally established residual can defeat the registered ordinary comparator without thereby deciding the measurement problem generally.	Preparation equivalence may remain open; a preparation side channel can block the target null; unresolved system-boundary, comparator, or empirical-entry burdens retain a not-yet-adjudicable posture.
APP-07 — Evolutionary Individuality	The tested lineage may retain higher-level selection capacity, endogenous individuating organization, life-cycle closure, or some combination of them after scaffold relaxation.	One or more account-specific criteria may fail while others survive; the intervention may show dependence on the external scaffold relation for a particular proposed form of individuality.	Evidence may remain insufficient for one or more account-specific questions. The experiment explicitly permits unresolved account states rather than requiring one definition to win.
APP-08 — Thermodynamic Arrow	A singular Arrow is conditionally available over a declared scope where the required physical asymmetry, realized-history provenance, relevant	No Lawful Translation where a required entry burden or PAA lineage	Not Yet Adjudicable where a required physical entry burden remains OPEN. The physical origin of the asymmetric

	orientation recognition, and scope-level whole-hold are warranted.	condition fails. A larger-scope singular Arrow can also fail through scope fracture while narrower Arrow partitions survive.	boundary may remain open even when the conditional Arrow relation closes.
APP-09 — Baryogenesis	Where the observed late asymmetry is retained and zero antecedence is independently established at scope S, GenerationFromZeroBurden(S) becomes live. Exhaustive establishment of zero across H _{adm} would close the stronger antecedent-zero burden at that scope.	One fully admissible history with genuinely nonzero net baryon number at the relevant antecedent boundary would defeat the universal all-history zero claim.	At corpus freeze, both the universal zero antecedent and a fully admitted nonzero-B counterexample at the same scope remain unestablished. The antecedent-zero question is therefore OPEN from both directions.
APP-10 — Turbulence	CLOSED WITHIN Σ_C is available if the compound DLBTF $\wedge$ HFN identity and its positive, negative, boundary, redundancy, and residual-identity burdens satisfy the frozen scientific protocol.	NOT CLOSED WITHIN Σ_C can follow from a scientifically valid necessity failure, sufficiency failure, redundancy result, residual-identity result, or other qualifying failure of the proposed compound identity.	NOT YET ADJUDICABLE WITHIN Σ_C applies where evidence or an applicable required burden remains unresolved. An inadequate assay does not count as an identity falsifier.
APP-11 — Cosmological Constant	A registered protection relation may PASS its perturbative or EFT update tests; other registered scientific jobs may independently close where their own burdens are satisfied.	A registered completion may FAIL its declared joint role without erasing an independently established parent protection result.	A whole-history burden may remain OPEN where the required complete-history information is unavailable. Required successor burdens may likewise remain OPEN rather than being relabeled as failures.

D.2 Non-binary structures beyond positive, negative, and OPEN

Application	Non-identification / comparator-relative result	Composite or mixed result structure	Multiple lawful outcomes / bounded conditionality
APP-01 — Failure Regimes	A candidate structural classification may tie an equally informed ordinary-physics comparator; such a tie does not establish incremental predictive utility.	Mixture and route heterogeneity can preserve more than one first-break contribution within one astrophysical object or probe aperture.	One first-break class need not map to one observable morphology; several physical realizations and observable subfamilies may follow from the same class. Conclusions remain bounded to the registered physical domain.

APP-02 — Consciousness	No separate non-identifying endpoint is required by the current contract.	Different candidate passages may simultaneously occupy POSITIVE and INDETERMINATE standings without being normalized into one global verdict.	Closure is conditional on obtaining the required two-sided experiential anchors at the declared bearer / passage scope; a realization mechanism may remain separately OPEN downstream.
APP-03 — Quantum Gravity	Failure to distinguish or exhaust the admitted classical comparison class blocks the stronger QG requirement rather than becoming evidence for it.	Individual experimental discriminator blocks can close while parent classical representability or the coverage bridge remains unresolved.	QGR is explicitly conditional on the conjunction of the three positive parent gates and GCR violation at one frozen scope.
APP-04 — Dark Energy	The proposed physical seat can remain empirically unidentified while the structural B-seat relation is well formed.	Structural contract closure and empirical physical identification occupy separate stages and may therefore carry different dispositions.	The physical-identification conclusion is conditional on all required E1–E4 entry burdens closing at the declared cosmological scope.
APP-05 — Life from Chemistry	Not a separate named endpoint; inability to classify continuity or maintenance remains INDETERMINATE rather than being converted into non-identification.	OCTT continuity class, AMC disposition, and later Life translation can lawfully differ. A continuity result survives even if a later maintenance or Life burden fails.	OCTT admits multiple scientifically meaningful continuity classes, including C1–C5. Life translation is conditional on an independently admitted AMC object and does not inherit success automatically.
APP-06 — Quantum Measurement	The experiment can locate an ordinary explanation—such as a non-Lüders instrument or finite-memory process—without identifying a novel measurement mechanism.	Preparation equivalence, side-channel status, fixed-instrument adequacy, process-memory adequacy, residual structure, and generalized-foil tests may close differently within the same program.	Each stronger result is conditional on prior entry and comparator burdens. Even the strongest residual establishes insufficiency of the registered ordinary comparator at the tested scope, not the identity of the ultimate replacement theory.
APP-07 — Evolutionary Individuality	No single definition is required to be selected by the experiment. Divergence among account-specific conclusions is itself informative.	The same lineage may satisfy higher-level selection and life-cycle closure while failing an endogenization criterion, or exhibit other account-specific combinations.	Convergence, divergence, partial retention, loss, resilience, and unresolved account states are all lawful within the same fixed lineage and scaffold challenge.
APP-08 — Thermodynamic Arrow	Failure to establish one global Arrow does not identify a unique alternative global structure.	A larger-scope claim can fail while narrower coherent Arrow partitions remain standing.	Multiple successor outcomes and lawful accounted return can remain compatible with the same PAA provenance constraint. The corollary therefore constrains lineage without selecting one unique future.

APP-09 — Baryogenesis	The scope result does not identify which admissible history was actually realized by the universe.	A model-level generation result, a history-class antecedent result, and actual cosmic-history identification remain separate scientific questions.	ObservedNonzero $\wedge$ ZeroEstablished(S) $\rightarrow$ GenerationFromZeroBurden(S) is explicitly scope-bounded and one-directional. No promotion from a model family to H_{adm} occurs for free.
APP-10 — Turbulence	No separate named non-identification endpoint is needed; unresolved discrimination remains NOT YET ADJUDICABLE rather than becoming candidate failure.	The released machine surface exposes located certificates—support, necessity failure, sufficiency failure, redundancy, residual identity, and others—rather than forcing every local statement into one result enum.	Final identity status is bounded to Σ_C and the frozen case roles. Local adverse results may coexist with other successfully adjudicated components while determining the appropriate global closure state.
APP-11 — Cosmological Constant	NOT IDENTIFIABLE is an explicit scientific endpoint where the mechanism is exactly degenerate with the frozen matched comparator on the tested channel.	The worked population simultaneously supports protection PASS + local NOT IDENTIFIABLE + whole-history OPEN + completion FAIL without allowing one endpoint to rewrite the others.	Protection, physical-successor discrimination, completion viability, and numerical realization are independently conditional on the jobs actually registered. A result at one layer does not automatically propagate to another.

D.3 Frozen corpus disposition and blocked stronger claim

Application	Present disposition at corpus freeze	Stronger result explicitly blocked
APP-01 — Failure Regimes	Pre-result demonstration protocol and discrimination architecture; no completed dark-matter execution or empirical confirmation claimed.	That the five classes have been empirically validated across dark matter generally, that every observable maps uniquely to one regime, or that Structural Flow adds predictive information beyond ordinary physics before comparator testing.
APP-02 — Consciousness	POSITIVE anchor admitted; no NEGATIVE anchor admitted; candidate absence cases remain INDETERMINATE; public translation is NotYetAdjudicable .	Empirical C3R placement, completed two-sided object closure, or a claim that absence of report, memory, behavior, or proxy detection establishes experiential absence.

APP-03 — Quantum Gravity	Structural experimental-adjudication architecture closed; empirical execution remains prospective; scoped QG requirement not yet established.	That gravity has already been shown to require a quantum description, that failure of one classical model exhausts the comparison class, or that a microscopic QG ontology has been selected.
APP-04 — Dark Energy	Structural / contract-shape burdens closed; required dark-energy empirical entry burdens remain OPEN; proposed physical identification is NotYetAdjudicable .	Empirical identification of dark energy with baseline completion B, microscopic origin, or exclusion of the comparison seats before the required evidence closes.
APP-05 — Life from Chemistry	Prospective contract; OCIT / AMC empirical states remain unresolved and the AMC → Life translation is not yet adjudicable.	That turnover continuity establishes autonomous maintenance, that autonomous maintenance establishes Life, or that assay inadequacy establishes absence of maintenance.
APP-06 — Quantum Measurement	Machine-clean Seam G / QCCT-I architecture; real domain-entry and target empirical burdens remain OPEN; no empirical QCCT result claimed.	Derivation of the Born rule, universal falsification of quantum mechanics, selection among interpretations, or identification of a new measurement mechanism merely from formal conformance.
APP-07 — Evolutionary Individuality	Machine-clean target and result architecture; scaffold-challenge experiment not performed.	A single universal definition of evolutionary individuality, generalization beyond the tested lineage / scaffold relation, or promotion of one account's result into the result of the others.
APP-08 — Thermodynamic Arrow	Conditional theoretical / formal corollary closed at declared scope; physical asymmetry and boundary conditions remain supplied by physics.	A universal Arrow across every region or branch, a unique future, absolute irreversibility, or derivation of thermodynamic entropy or its physical origin from PAA.

APP-09 — Baryogenesis	Generation-from-zero scope relation formalized; all-history zero antecedent remains OPEN, and no fully admitted nonzero-B history at the same comparison boundary has been established.	That the observed late asymmetry itself proves a zero antecedent, that a model-family premise applies to H_{adm} , or that the actual cosmic history has been identified.
APP-10 — Turbulence	Extensive prospective identity/adjudication architecture; empirical closure within Σ_C has not been executed.	That $DLBTF \wedge HFN$ is already the empirical identity of turbulence, that blocked assays count as falsifiers, or that the candidate is complete before redundancy and residual-identity burdens are adjudicated.
APP-11 — Cosmological Constant	Worked cross-update case: protection PASS , local channels NOT IDENTIFIABLE , direct whole-history burden OPEN , registered completion FAIL .	That the mechanism has thereby “solved the cosmological-constant problem,” that local protection establishes empirical identification or numerical realization, or that failure of one completion erases the parent protection result.

D.4 TERM-01 — Morphogenesis / Regeneration Scaling

TERM-01 was stopped before empirical execution, so it contributes **prospective result grammar rather than an observed scientific disposition**.

The architecture distinguished at least three relevant branches. A positive scale-conditioned result would require retained morphogenetic competence together with evidence that regenerated extent lawfully tracks the registered scale relation. A bounded negative scaling result would require patterning competence to survive while proportional scaling fails. If the intervention destroyed or rendered unassessable the patterning competence required to test scaling, the scaling question would remain OPEN rather than being converted into a negative result.

The prospective architecture also allowed donor-associated and host-associated scale dependence to separate. Those branches could therefore have yielded distinct causal interpretations without requiring every readout to rise and fall together.

Because the vertebrate protocol was not frozen or executed, none of those branches is an empirical result in the present corpus.

D.5 Comparative result

The matrices expose three recurring properties of the frozen corpus.

First, **negative evidence is not defined by absence of positive closure**. Consciousness requires a dedicated NEGATIVE route; Life / AMC requires a warranted substantive violation; turbulence distinguishes inadequate adjudication from identity failure; and dark-energy translation distinguishes OPEN from violated entry burdens.

Second, **scientific jobs that have been separated upstream can retain different outcomes downstream**. Continuity can survive while maintenance fails; a protection relation can pass while empirical identification fails or remains unavailable; a larger-scope Arrow can fracture while narrower Arrows survive; and one evolutionary-individuality criterion can fail while another remains satisfied.

Third, **strong structural constraint does not require one terminal scientific future**. The Arrow program permits multiple lawful successors under the same provenance constraint; baryogenesis makes the generation-from-zero burden conditional on an independently established antecedent; quantum measurement permits successively stronger comparator closures; and the cosmological-constant program permits simultaneous PASS, NOT IDENTIFIABLE, OPEN, and FAIL results because those states answer different registered questions.

The corpus therefore does not support a model in which formalization simply converts heterogeneous inquiries into one binary success criterion. Its stronger commonality is narrower:

a result is licensed only for the burden, object, level, and scope that have actually been answered.

OPEN is not FALSE.

Non-identification is not generic failure.

Failure downstream does not automatically erase an earned upstream result.

A bounded positive does not inherit adjacent claims for free.

Appendix E. Threat and Falsifier Matrix

The adversarial audit distinguishes attacks that would directly change the status of the Universal Kernel from attacks that would instead defeat an application, weaken the interpretation of kernel stability, or narrow the paper’s evidentiary claim.

That distinction is necessary because not every failure in a kernel-mediated scientific program is a kernel failure. A false empirical premise belongs first to the scientific layer. A distorted translation belongs first to the semantic interface. A validly encoded record that receives an invalid formal license reaches the kernel itself.

The matrices below therefore ask, for each attack: **what exactly must fail, what evidence would count, what superficially similar result would not count, and what consequence would follow if the attack succeeded?**

E.1 Direct attacks on kernel status or the historical non-revision claim

Falsifier / threat	Exact attack	What would count / what would not count	Current status and consequence
Archival identity failure	Show that one or more application packages used a materially different Universal Kernel while the study treats the release series as carrying the same frozen file.	Counts: archive evidence showing a different kernel file or non-matching archived checksum in a package included in the non-movement claim. Does not count: different adapter files, companions, scientific anchors, local contracts, filenames elsewhere in the package, or a later kernel version outside the frozen corpus.	Current status: not found; Appendix F records the common archive identity used by the study. If demonstrated: the historical file non-movement claim must be corrected or restricted. It would not, by itself, show that either kernel version is scientifically or formally inadequate.
Hidden universal repair / adapter shadowing	Show that an application remained nominally on the frozen kernel only because a necessary kernel-owned structural job was silently recreated, altered, or repaired in application-specific machinery.	Counts: a local adapter carries a necessary, reusable, domain-neutral structural burden absent from the kernel, and the application could not lawfully perform its declared adjudication without that burden. Also counts if local code silently changes the meaning of a universal requirement while presenting the result as ordinary reuse. Does not count: domain mechanisms, assays, comparators, empirical predicates, evidence rules, or other scientifically local burdens that can honestly remain local.	Current status: no qualifying case located. Adapter elasticity remains a live threat. If demonstrated: the affected non-revision result would be reopened; the burden would require authority-level adjudication and could earn kernel revision if it satisfies the universal-placement test.
Missing reusable universal capacity / coherent-object exclusion	Produce an independently defensible scientific object whose lawful adjudication requires a structural capacity the current kernel cannot represent faithfully.	Counts: the missing capacity is necessary for the object, performs a located structural job, has a located capacity loss if absent, is reusable and domain-neutral, cannot be reduced to existing universal structure, and cannot honestly remain object-specific or domain-owned. Does not count: a new scientific mechanism, assay, comparator, measurement convention, local	Current status: no such burden located in the frozen corpus. If demonstrated: this is genuine kernel revision pressure. The historical non-revision result would remain true for the frozen corpus, but the current kernel would no longer be sufficient for the demonstrated broader scope.

		evidentiary rule, or unfamiliar domain detail merely absent from the kernel.	
Faithful-record false conformance	Give the kernel a faithfully populated record containing a required kernel-owned violation or unresolved burden and obtain structural conformance anyway.	Counts: the contradictory, VIOLATED, or OPEN required burden is explicitly and faithfully represented in the record the kernel actually receives, lies within a burden the kernel claims to govern, and the formal machinery nevertheless licenses the prohibited conformance. Does not count: hidden contradictions, invented evidence, omitted predicates, dishonest witnesses, malformed adapters, or scientific facts never encoded into the inspected record.	Current status: no within-scope false-conformance counterexample located. If demonstrated: direct counterexample to the kernel's adjudication logic at its claimed authority. Formal repair would be required.
Self-support / conclusion used as its own prerequisite	Show that a desired conclusion, or an equivalent restatement of it, is permitted to establish one of the premises required to license that same conclusion.	Counts: the target scientific identification, closure, object, or success state supplies its own translation warrant, object admission, evidence predicate, comparison result, or other required antecedent without independent support. Does not count: independently warranted evidence that happens to support the eventual conclusion, or a downstream conclusion lawfully derived after its antecedents close.	Current status: no qualifying circular route established in the frozen audit; non-self-support is an explicit common admission burden. If demonstrated in an application: that application's claimed closure is invalid. If the universal contract itself licenses the faithfully encoded circular route: the failure escalates to kernel-level false conformance.
OPEN upgraded into positive license	Show that a required unresolved entry burden can remain OPEN while the same claim is nevertheless licensed as structurally conforming or positively established.	Counts: a required universal, object-specific, or domain-entry burden is faithfully encoded as OPEN and the formal system still licenses the result whose admission requires its discharge. Does not count: a downstream-domain burden remaining OPEN after a narrower structural translation has already lawfully conformed; nor does a human overstatement of a correctly returned NotYetAdjudicable state constitute a kernel defect.	Current status: no such machine route located; several applications explicitly stop at OPEN / NotYetAdjudicable. If demonstrated: direct failure of the common gating architecture and therefore kernel-level repair pressure.
Universal non-guarantee failure	Show that the universal machinery itself supplies or preselects a scientific outcome that the governing domain was supposed to adjudicate.	Counts: a preferred identity, mechanism, physical result, empirical classification, unique continuation, or stronger closure follows merely from universal premises while a necessary domain-owned discriminating burden remains unresolved or absent. Does not count: a strong conditional result obtained after independently warranted scientific premises and local burdens have been supplied.	Current status: no qualifying outcome-preselection route found in the audited corpus. If demonstrated: the central methodological architecture of the paper fails at that point; the offending universal premise or inference would require removal, restriction, or rederivation.

E.2 Attacks on interpretation, application validity, and transfer

Threat	Exact attack	What would count / what would not count	Current status and consequence
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Semantic mispopulation / authority leakage	Show that a formally clean record materially misrepresents the scientific object, evidence state, comparator, scope, or domain-owned premise while the machine result is nevertheless presented as scientific warrant.	Counts: scientifically false, misleading, or inadequately warranted predicates are entered as discharged; a relevant distinction is mistranslated; or machine conformance is used as evidence that the translation itself was scientifically faithful. Does not count: an honestly represented scientific disagreement or uncertainty that remains visible as OPEN or domain-owned.	Current status: major surviving limitation by design. If demonstrated: the affected scientific conclusion loses warrant, but the result is not automatically a kernel falsifier because the kernel does not claim to establish the truth or faithfulness of its supplied scientific record.
Independently reproduced object-disambiguation incompatibility	An independent investigator derives a materially different but scientifically defensible object or burden partition for the same problem and scope, and the rival decomposition survives comparable failure, evidence, scope, and non-circularity pressures.	Counts: rival objects remain independently admissible yet support materially incompatible adjudications that cannot be resolved merely by identifying different scopes or scientific jobs. Does not count: alternative terminology, harmless redescription, decompositions at different levels or scopes, or rivals for which ordinary domain evidence can discriminate.	Current status: not yet tested; decomposition uniqueness is not claimed. If demonstrated: the paper's object-disambiguation interpretation must be narrowed or extended. If the kernel cannot faithfully represent one independently defensible rival because a reusable structural capacity is missing, the case may escalate to coherent-object exclusion.
Theorem-coverage overstatement	Treat inclusion of the full kernel, semantic reliance on a distinction, or upstream use during derivation as though it were direct executable application coverage.	Counts: claiming that a theorem family survived direct cross-domain stress when the source audit did not establish application-adapter use. Does not count: explicitly distinguishing DIRECT, GENERIC / DEPENDENCY, SEMANTIC ONLY, and NONE ESTABLISHED.	Current status: controlled by the source audit and Appendix B; coverage remains explicitly uneven. If demonstrated: the manuscript's validation claim must be corrected. It would not show that the untested theorem family is false; it would show only that its testing had been overstated.
Post-result object / scope / protocol mutation without reopening	Preserve a preferred result by changing the object, level, scope, comparator, test role, protocol, or evidentiary meaning after the relevant outcome is known while treating the revised claim as the original frozen test.	Counts: a materially result-sensitive change alters what was being tested or what would have counted as failure without reopening the claim, registering an amendment, or treating the new analysis separately. Does not count: prospectively declared branches, pre-result amendments with recorded lineage, or an explicit post-result reopening that forfeits the original confirmatory claim.	Current status: no corpus-wide defeating case established; multiple packages contain scope freezes, fixed comparators, prospective branches, or reopening rules. If demonstrated: the affected result loses its confirmatory standing and must be reclassified as exploratory, reopened, or newly scoped.
Independent-transfer failure	External investigators cannot use the frozen architecture without author-dependent reconstruction, hidden	Counts: serious independent applications repeatedly require undocumented semantic decisions, private scaffolding, or kernel alteration	Current status: not yet tested. No independent full external application population exists. If demonstrated: the paper's within-program historical

	interpretation, or ad hoc modification of the universal machinery.	merely to reproduce the claimed object-to-contract workflow. Does not count: disagreement with a scientific premise, unsuccessful experiments, unfamiliarity with Structural Flow terminology, or an independently located kernel revision pressure.	result survives, but claims about practical transferability or general methodological reuse would need substantial narrowing.
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E.3 Live scope-pressure candidate: pairwise → n-ary Coherence

The audit identified one especially clean future pressure surface that is **not a present falsifier**.

Pressure candidate	Exact attack	What would count / what would not count	Current status and consequence
Pairwise → n-ary / global Coherence	Construct a scientifically legitimate case with three or more locally admissible structures in which the currently licensed pairwise relations all succeed, while no simultaneous global realization exists. Then ask whether the missing global relation performs a reusable universal job.	Counts as revision pressure only if: the global whole-hold relation is necessary to the scientific adjudication; pairwise composition cannot honestly represent it; the missing job and capacity loss are located; the relation is reusable and domain-neutral; and it cannot honestly remain local. Does not count as a current falsifier: merely exhibiting global frustration beyond the explicitly pairwise scope of the released theorem. Pairwise success does not presently claim arbitrary n-ary success.	Current status: live future pressure candidate. If the stronger conditions are met: a justified n-ary/global extension of universal Coherence may be earned. The existing pairwise theorem would not thereby become false; its declared scope would remain intact.

E.4 Adjudication rule

The attack matrices preserve four different outcomes that should not be collapsed.

A defect in **scientific premises** invalidates or reopens the scientific application.

A defect in **semantic translation** invalidates or reopens the adapter or contract.

A faithfully represented defect in a relation the **kernel itself claims to govern**, followed by an invalid formal license, reaches the Universal Kernel directly.

A necessary reusable structural capacity that the current kernel cannot faithfully carry creates **revision pressure without requiring that an existing theorem have been false**.

Accordingly, the following do **not** by themselves falsify the Universal Kernel:

- an unsuccessful experiment;
- a scientifically false domain hypothesis;
- an unresolved burden;
- a local mechanism absent from universal code;
- a malformed or dishonest adapter;
- an unencoded contradiction;
- failure of one application-specific candidate;
- proof-engineering difficulty;
- or an out-of-scope demand placed on a theorem whose declared scope is narrower.

The governing test is instead:

Locate the failed job, the level at which it fails, the capacity lost, and the lowest authority capable of repairing it honestly.

E.5 Current adversarial disposition

Across the frozen audited corpus:

No archival identity failure has been established.

No hidden universal repair sufficient to earn kernel revision has been established.

No necessary reusable domain-neutral capacity absent from the current kernel has been demonstrated.

No faithfully represented required kernel-owned VIOLATED or OPEN burden has been found to receive false structural conformance.

No universal premise has been shown to preselect the scientific outcome in place of the governing domain evidence.

Several threats nevertheless remain live: semantic faithfulness, adapter elasticity, decomposition uniqueness, uneven theorem-family coverage, external transfer, and the possibility that future multi-component applications will expose a genuine n-ary/global Coherence burden.

The present disposition is therefore:

NO CURRENT KERNEL REVISION EARNED.

That is a status of the frozen audit, not a guarantee against future defeat.

Appendix F. Kernel Checksum and Archive Manifest

This appendix records the archival file-identity evidence used for the paper’s historical kernel non-movement claim.

The reference kernel is:

Structural_Flow_Universal_Kernel_v1.0.lean

The Zenodo-recorded checksum used for cross-package comparison is:

MD5: 4fe6d868650766eba368891ba008ae4d

MD5 is used because it is the checksum recorded by Zenodo for the archived kernel copies in the audited release packages. The manifest therefore reports the archival evidence actually available across those releases rather than substituting a different locally computed hash.

F.1 DOI-to-kernel archive manifest

For every row below, **REFERENCE MATCH** means that the archived package contains **Structural_Flow_Universal_Kernel_v1.0.lean** with Zenodo-recorded MD5 **4fe6d868650766eba368891ba008ae4d**.

Application	Version-specific DOI	Archive identity
APP-01	10.5281/zenodo.23084141	REFERENCE MATCH
APP-02	10.5281/zenodo.22812662	REFERENCE MATCH
APP-03	10.5281/zenodo.22833457	REFERENCE MATCH

APP-04	10.5281/zenodo.22845714	REFERENCE MATCH
APP-05	10.5281/zenodo.22848429	REFERENCE MATCH
APP-06	10.5281/zenodo.22879295	REFERENCE MATCH
APP-07	10.5281/zenodo.22907164	REFERENCE MATCH
APP-08	10.5281/zenodo.22948809	REFERENCE MATCH
APP-09	10.5281/zenodo.22968518	REFERENCE MATCH
APP-10	10.5281/zenodo.23026122	REFERENCE MATCH
APP-11	10.5281/zenodo.23174925	REFERENCE MATCH

Across all eleven version-specific DOI packages in the frozen public corpus, the archived kernel therefore matches the same reference file and Zenodo-recorded MD5.

The manifest supports the historical statement used in the main text:

the kernel file did not move across the audited release corpus.

This result is limited to file identity. Checksum agreement does not establish that the kernel is formally complete, scientifically correct, sufficient for every application, minimal, optimal, or immune to future revision. It also does not establish that every theorem family in the file was exercised by every application; direct theorem-family coverage is reported separately in Appendix B.

F.2 Supplemental resident checksum

The resident audit copy of the reference kernel has the following independently computed checksum:

SHA-256:

b455ec869d062f0fc05d239c5006c843165cd598edbd8c35eae442ff471689a3

This SHA-256 value is reported only as a supplemental checksum of the resident audit copy. It is **not** represented as a Zenodo-recorded cross-package checksum because equivalent archive-level SHA-256 records were not available for every package in the audited release series.

The archival cross-package identity claim in this study therefore rests on the **Zenodo-recorded MD5 manifest above**, not on an inferred or retrospectively substituted SHA-256 manifest.

F.3 Evidentiary ceiling

The manifest answers one narrow historical question:

Did the archived Universal Kernel file change across the eleven audited DOI packages?

On the archive-recorded checksum evidence used in this study, the answer is **no**.

It does not answer the stronger question:

Was that unchanged kernel adequate, correct, complete, or universally sufficient?

Those are methodological and scientific questions addressed by the source audit, cross-application comparison, adversarial analysis, and future falsification routes—not by checksum identity.

Appendix G. Morphogenesis / Regeneration Scaling — Terminated-Case Note

G.1 Classification

Morphogenesis / Regeneration Scaling is retained in this study as **TERM-01**, a separately classified architectural stress case.

The program was:

terminated after substantial causal experiment architecture, before protocol freeze and empirical execution.

TERM-01 is not one of the eleven completed/source-audited public application packages and is not included in $N = 11$. It has no public DOI or public source package asserted in this study.

Its inclusion serves a narrow methodological purpose: the case progressed far enough to test object disambiguation, prospective causal architecture, result routing, and possible pressure on the Universal Kernel, but it was deliberately stopped before the proposed vertebrate intervention was frozen or performed.

G.2 Scientific object

The broad problem concerned how regenerating biological systems preserve scale-relative form as the size relevant to the regenerative process changes.

The retained scientific object was **scale-conditioned morphogenesis**:

a morphogenetic system may preserve scale-relative pattern or growth as the spatial extent relevant to that process changes.

The object was deliberately narrower than a general theory of morphogenesis, regeneration, positional information, or organismal size control.

G.3 Critical discrimination

The central discrimination was:

patterning survives; proportional scaling fails.

This distinction was necessary because a manipulation that destroys general regenerative or patterning competence cannot lawfully be treated as evidence against proportional scaling itself. A negative scale-relation result requires the relevant morphogenetic competence to remain sufficiently intact for the scale question still to be answerable.

Accordingly:

loss of patterning competence $\neq$ negative evidence for proportional scaling.

That firewall was retained throughout the derived experimental architecture.

G.4 Architecture reached

Before termination, the project reached the following causal architecture:

- retained patterning competence was separated from proportional scaling;
- a candidate vertebrate regeneration system was selected;
- local retained information was separated from host- or extra-local scale context;
- a reciprocal transplantation architecture was derived to place those causal possibilities under contrast;
- local and host-associated scale coordinates were to be fixed prospectively rather than reconstructed after the regenerative outcome;
- nested signaling and final-growth readouts were planned;
- prospective result branches distinguished donor/local dominance, host-associated dominance, joint contribution, signaling-only effects, absence of the predicted relation, and failure of morphogenetic competence.

The resulting design was intended to distinguish whether regenerated extent answered primarily to information retained with the transplanted tissue, to the surrounding host scale, to both, or to neither while preserving the competence firewall above.

The architecture reached a substantially causal experiment rather than a descriptive correlation study. It did not, however, reach protocol freeze.

G.5 Stop point

Development stopped before:

- final vertebrate protocol freeze;
- finalized execution parameters;
- animal intervention;
- empirical data collection;
- or result adjudication.

The project was deliberately not advanced after review of the ethical and welfare cost of the proposed vertebrate experiment relative to the expected scientific gain.

The stop is therefore part of the methodological record rather than an experimental outcome.

G.6 Formal status

At the point of termination, the scoped formal architecture was judged adequate for the scientific object that had been reached.

No reusable, domain-neutral structural burden had been exposed that required amendment of the frozen Universal Kernel.

That statement has a strict ceiling. It means only that **no kernel-repair burden had appeared before the program was stopped**. It does not establish what later protocol development, execution, specialist criticism, or empirical results might have exposed.

G.7 Claim ceiling

TERM-01 contributes no empirical evidence to the present paper.

This appendix does **not** claim:

- that proportional scaling is governed by the proposed causal architecture;
- that donor-associated or host-associated scale information would have dominated;
- that the proposed experiment would have produced interpretable data;
- that the experiment would have succeeded;
- that the proposed scientific object is unique;
- or that the formal architecture would necessarily have survived later execution.

No unpublished machine filename is part of the public claim, and no public source package or DOI is asserted.

The lawful methodological statement is narrower:

Morphogenesis reached an experimentally answerable residual scientific object and a substantially derived causal experimental architecture without exposing a Universal-Kernel repair burden before ethical termination.

Appendix H. Reproducibility and Public Source-Package Index

This appendix is navigational.

Scientific claims remain governed by the cited public artifacts and their archived package contents. The index does not transfer scientific authority from an anchor artifact to its companion, adapter, contract, or Universal Kernel.

For the eleven public applications, the version-specific Zenodo DOI is the primary archival retrieval point used by this study. Each DOI identifies one coordinated scholarly release package rather than a collection of independent publications. The machine companion and scoped formal files are supporting components of the corresponding release and do not acquire independent scientific authority merely by inclusion.

The public packages use the same frozen universal file:

Structural_Flow_Universal_Kernel_v1.0.lean

Baryogenesis is the one three-layer package in the corpus: it contains no separate human-readable machine companion. The other ten public application packages use a four-layer architecture.

H.1 Public scientific anchors and archival retrieval

ID	Public scientific anchor	Version-specific DOI
APP-01	<i>Five Failure Regimes in Dark Matter Light-Probes</i> v2.1	10.5281/zenodo.23084141
APP-02	<i>Consciousness and Experiential Absence</i> v1.7	10.5281/zenodo.22812662
APP-03	<i>When Is Quantum Gravity Required?</i> v1.4	10.5281/zenodo.22833457

APP-04	<i>Where Does the Dark-Energy Contribution Sit?</i> v1.0	10.5281/zenodo.22845714
APP-05	<i>Life from Chemistry</i> v1.2	10.5281/zenodo.22848429
APP-06	<i>Quantum Measurement Anchor Artifact</i> v1.0	10.5281/zenodo.22879295
APP-07	<i>Evolutionary Individuality as an Empirical Question</i> v1.0	10.5281/zenodo.22907164
APP-08	<i>The Thermodynamic Arrow</i> v1.5	10.5281/zenodo.22948809
APP-09	<i>The Missing Premise in Baryogenesis</i> v0.9	10.5281/zenodo.22968518
APP-10	<i>The Turbulence Identity Problem</i> v1.2	10.5281/zenodo.23026122
APP-11	<i>Stress Tests for Cosmological-Constant Mechanisms</i> v1.7	10.5281/zenodo.23174925

The DOI package is sufficient as the archival public-source location for the purposes of this paper. Where a corresponding public source mirror is maintained elsewhere, that mirror is supplementary; this index does not rely on a separate mirror to establish the audited package identity.

H.2 Public machine and contract components

ID	Human-readable machine / execution companion	Public scoped formal component

APP-01	<i>Five Failure Regimes Light Probe Experiment Companion</i> v0.5	Dark_Matter_Failure_Regime_Adapter_Contract_v0.6.lean — domain adapter / contract
APP-02	<i>Consciousness and Experiential Absence Machine Companion</i> v0.5	SF_Consciousness_Contract_and_Adapter_v0_1.lean — scoped contract / adapter
APP-03	<i>Quantum Gravity Adjudication — Machine Companion</i> v1.3	Quantum_Gravity_Adjudication_Formal_Contract_v1.0.lean — scoped formal contract
APP-04	<i>Dark Energy Contract Companion</i> v0.4	Dark_Energy_Baseline_Completion_Contract_Append_v0.1.lean — scoped Lean contract
APP-05	<i>Life from Chemistry Contract Adapter Companion</i> v0.4	SF_Life-Origin_of_Life_Contract_and_Adapter_v0_5.lean — scoped Life / Origin-of-Life contract and adapter
APP-06	<i>Quantum Measurement: QCCT Audit and Execution Companion</i> v1.5	SF_Quantum_Measurement_Contracts_and_Adapters_v1_0.lean — consolidated Seam G / Born and QCCT-I contracts and adapters
APP-07	<i>Evolutionary Individuality Adapter_Contract Companion</i> v1.1	Evolutionary_Individuality_Domain_Adapter_Contract_v0_2.lean — scoped domain adapter / contract
APP-08	<i>SF Arrow Corollary Contract Companion</i> v1.1	SF_Arrow_Corollary_Contract_and_Adapter_v0_3.lean — scoped contract / adapter
APP-09	No separate public companion	Baryogenesis_Scope_Formal_Contract_v0.1.lean — scoped formal contract / adapter

APP-10	<i>Turbulence Identity Companion</i> v1.3	<code>Turbulence_DLBTF_HFN_Compound_Identity_APPEND_BUNDLE_v1.0.lean</code> — turbulence-specific adapter / contract bundle
APP-11	<i>CCP Contract and Adapter</i> <i>Companion</i> v0.7	<code>CCP_Stability_Routing_Adapter_v0.7_CONSOLIDATED.lean</code> — CCP-specific routing adapter / contract

H.3 Authority and use of the package components

The components above perform different jobs and should not be treated as interchangeable sources.

The **scientific anchor** governs the scientific object, scientific claims, empirical interpretation, evidentiary posture, and claim ceiling.

The **human-readable companion**, where present, explains the machine architecture and the relation between the scientific anchor, scoped contract, adapter, and Universal Kernel. It is an audit and interpretation surface rather than an independent scientific authority.

The **scoped formal component** encodes the application-specific contract, adapter, burdens, dispositions, blockers, certificates, or routing logic used by the machine layer. In several releases the contract and adapter are intentionally combined in one Lean file rather than distributed as separate public files.

The **Universal Kernel** supplies the frozen reusable formal interface against which the scoped machinery is checked. Its presence does not transfer scientific authority from the governing domain to Lean.

Accordingly, the intended audit route for an external reader is:

scientific anchor → **companion where present** → **scoped formal component** →
frozen Universal Kernel

rather than the reverse.

A reader investigating a scientific premise should return to the scientific anchor and its cited domain evidence. A reader investigating the semantic translation should inspect the anchor together with the companion and scoped formal component. A reader investigating whether the stated formal consequence follows from the supplied record should inspect the scoped Lean component and the compatible Universal Kernel.

H.4 Corpus boundary

TERM-01, Morphogenesis / Regeneration Scaling, does not appear in this public source-package index.

It is an unpublished retained project case, has no public DOI or public source package asserted in this study, and is represented only through the bounded methodological note in Appendix G.

Internal audit spines, maintenance trackers, governance files, working derivation materials, and unpublished TERM-01 machine materials likewise do not belong in this public reproducibility index.

H.5 Reproducibility ceiling

The index makes the frozen application record publicly inspectable. It permits an outside reader to retrieve the scientific anchors, inspect the coordinated formal materials, compare package architecture, and verify the common frozen-kernel identity reported in Appendix F.

It does not establish independent replication.

No independent specialist has yet selected a new scientific problem, independently formed its object and translation, and carried a full external application against the frozen kernel. Public inspectability is a prerequisite for that test, not a substitute for it.

AI Use Disclosure

Artificial intelligence tools were used during the development of this work for research assistance, source auditing, adversarial review, cross-application comparison, conceptual organization, drafting support, editing, consistency checking, and formalization support. The author directed the research program, selected and froze the comparative corpus, determined the audit architecture and authority boundaries, evaluated the cross-domain findings and limitations, and takes responsibility for the final content.

From Non-Guarantee to Answerability

Cross-Domain Stress Tests of a Universal Formal Kernel

Signed,

A handwritten signature in black ink, consisting of a large, stylized 'R' followed by a smaller, more fluid signature.

Rick